

—* Documentation section *—

beginDocumentation()

```
document {
Key => NongeneralTypeSurfacesInP4,
Headline => "Construction of smooth non-general type surfaces in P4",
  "In 1989 Elligsrud and Peskine proved a conjecture of Harstshorne and Lichtenbaum about smooth rational surfaces in P4. In fact,
more generally:
  There are only finitely many components of the Hilbert scheme of surfaces in P4, whose general point correspond to a smooth
  surface not of general type.
```

```

  In that period there was a flourish of activities to construct such surfaces, in part using Computer Algebra. In this package we
review
  those constructions, which after 30 years of Macaulay2 have become simpler and faster. Moreover we have discovered a few
further
  examples",
```

```

  PARA{},
  SUBSECTION "Rational surfaces",
  UL{
    TO unirationalFamiliesOfRationalSurfaces,
    TO schreyerSurfaces,
    TO aboRanestadSurfaces
  },
  SUBSECTION "Enriques and K3 surfaces",
  UL{
    TO enriquesSurfaceOfDegree9,
    TO enriquesSurfaceOfDegree10,
    TO enriquesSurfaceD11S10,
    TO K3surfaces,
  },
  SUBSECTION "Irregular surface",
  UL{
    TO quinticEllipticScroll,
    TO ellipticConicBundle,
    TO horrocksMumfordSurface,
    TO abelianSurfaceD15,
    TO abelianSurfaceD15S21Popescu,
    TO biellipticSurfaceD10,
    TO biellipticSurfaceD15,
    TO irregularEllipticSurfaceD12,
  },
  SUBSECTION "Elliptic surfaces",
  UL{
    TO irregularEllipticSurfaceD12,
    TO surfacesOfKodairaDimension1,
  },
  SUBSECTION "Investigating embedded surfaces",
  UL{
    TO adjunctionProcessData,
    TO residualInQuintics,
    TO canonicalDivisor,
    TO selfIntersectionNumber,
    TO tateResolutionOfSurface,
    TO numericalFunctions,
  }
}
```

```
document {
Key => unirationalFamiliesOfRationalSurfaces,
Headline => "unirational families of rational surfaces",
  "Most of the families constructed in [DES], [Popescu1] and before are actually unirational. We list the link to the corresponding
functions.
  An exception are certain families of Schreyer and Abo–Ranestad surfaces, where we only know that some of the families are
unirational.",
```

```

  PARA{},
  SUBSECTION "non-degenerate rational surfaces in P4",
```

```

    UL{
      TO cubicScroll,
      TO delPezzoSurface,
      TO castelnuovoSurface,
      TO bordigaSurface,
      TO ionescuOkonekSurfaceD7,
      TO okonekSurfaceD8S6,
      TO ionescuOkonekSurfaceD8S5,
      TO nonspecialAlexanderSurface,--
      TO specialityOneAlexanderSurface,--
      TO degree10DESSurface,--
      TO degree10pi9RanestadSurface,--
      TO degree10pi8RanestadSurface,--
      TO popescuSurface,
      TO vBELSurface
    },
    PARA{},
    SUBSECTION "further families",
    UL{
      TO schreyerSurfaces,
      TO aboRanestadSurfaces
    }
  }
}
document {
  Key => numericalFunctions,
  Headline => "Various numerical functions to investigate surfaces in P4",

```

```

  PARA{},
  SUBSECTION "Intersection numbers",
  UL{
    TO LeBarzN6,
    TO Ksquare,
    TO HdotK,
    TO sectionalGenus,
  },

  SUBSECTION "Tate resolutions",
  UL{
    TO tateResolutionOfSurface,
    TO chilTable,
    TO chiO,
    TO irregularity,
    TO geometricGenus,
  },

  SUBSECTION "6-secants and canonical divisors",
  UL{
    TO canonicalDivisor,
    TO partitionOfCanonicalDivisorOfAboSurface,
    TO residualInQuintics,
  },

```

```

}

```

```

document {
  Key => schreyerSurfaces,
  Headline => "functions concerning Schreyer surfaces (9 or 10 families)",
  "[Schreyer,1996] discovered 4 families of surfaces X in P4 with d=11, sectional genus pi=10 via a search over a finite field
  of which 3 families consist of rational surfaces.
  Repeating such search now, we found altogether 10 families. In the following we give an overview
  of the functions used in that search.",
  PARA{},
  SUBSECTION "from modules to surfaces",
  UL{
    TO schreyerSurfaceFromModule,
    TO moduleFromSchreyerSurface,
    TO exampleOfSchreyerSurfaces,

```

```

    TO specificSchreyerSurface
  },

  SUBSECTION "search for modules",
  UL{
    TO findRandomSmoothSchreyerSurface,
    TO collectSchreyerSurfaces
  },

  SUBSECTION "lift to characteristic zero and unirational or nearly unirational families",
  UL{
    TO tangentDimension,
    TO specialEnriquesSchreyerSurface,
    TO schreyerSurfaceWith2LinearSyzygies,
    TO schreyerSurfaceWith2or3LinearSyzygies,
  }
}

document {
  Key => aboRanestadSurfaces,
  Headline => "functions concerning Abo-Ranestad surfaces ( 9 families)",
  "[Abo-Ranestad,199x] discovered 4 families of rational surfaces X in P4 with d=12, sectional genus pi=13 via a search over a finite field.
  Reviewing their construction we found altogether 9 families.
  Most of these components are unirational.",

  PARA{},
  SUBSECTION "from matrices to surfaces",
  UL{
    TO aboRanestadSurfaceFromMatrix,
    TO matrixFromAboRanestadSurface
  },

  SUBSECTION "search for modules",
  UL{
    TO aboRanestadSurface,
    TO specificAboRanestadSurface,
    TO get4x2Matrix,
    --TO singAboRanestadSurfacesStatistic,
  },

  SUBSECTION "lift to characteristic zero",
  UL{
    --TO tangentComputation,
    TO veroneseImagesInG25
  }
}

document {
  Key => K3surfaces,
  Headline => "Known families of K3 surfaces",
  "Various family of non-minimal K3 surfaces are known.
  We enumerate the families by the degree D, the sectional genus S and
  their 6-secant behaviour L. Note that a smooth surface in P4 is expected to have
  finitely many 6-secants. If this number is finite, then Le Barz 6-secant formula
  computes sum of the number of 6-secants and the number of  $(-1)$  lines on the surface.
  Since every 6-secant line is contained in the zero locus of the ideal generated by the quintics
  containing the surface, the number of 6-secant can often be determined from the equation
  of the surface, hence in that case we get information about the  $(-1)$ -Lines, ie. ,
  the curve contacted by the first adjunction map.",

  PARA{},
  SUBSECTION "K3 surfaces",
  UL{
    TO K3surfaceD6,
    TO K3surfaceD7,
    TO K3surfaceD8,
    TO K3surfaceD9,
    TO K3surfaceD10S9L1,
  }
}

```

```

    TO K3surfaceD10S9L3,
    -- TO H1module,
    TO K3surfaceD11S11Ln,
    TO K3surfaceD11S12,
    TO K3surfaceD12,
    TO K3surfaceD13,
    TO K3surfaceD14,
    TO aboSurfaces,
  },
  PARA{},
  SUBSECTION "6-secants and adjunction",
  UL{
    TO LeBarzN6,
    TO residualInQuintics,
    TO canonicalDivisor,
    TO partitionOfCanonicalDivisorOfAboSurface,
    TO selfIntersectionNumber,
  },
}

document {
  Key => aboSurfaces,
  Headline => "functions for investigating Abo surfaces, (9 families)",
  "",

  PARA{},
  SUBSECTION "K3 surfaces of degree 12 and sectional genus 13",
  UL{
    TO aboSurfaceFromMatrix,
    TO testMatrix1,
    TO testMatrix2,
    TO randomAboSurface,
    TO analyzeAboSurface,
    TO collectAboSurfaces,
    TO specificAboSurface,
  },
  PARA{},
  SUBSECTION "unirational or nearly unirational constructions",
  UL{
    TO abo111333Surface,
    TO abo111117Surface,
  },
  PARA{},
  SUBSECTION "6-secants and adjunction",
  UL{
    TO LeBarzN6,
    TO residualInQuintics,
    TO canonicalDivisor,
    TO partitionOfCanonicalDivisorOfAboSurface,
    TO selfIntersectionNumber,
  },
}

document {
  Key => surfacesOfKodairaDimension1,
  Headline => "surface of Kodaira dimension 1 (10 families)",
  "",

  PARA{},
  SUBSECTION "elliptic surfaces",
  UL{
    TO ellipticSurfaceD7,
    TO ellipticSurfaceD8,
    TO ellipticSurfaceD9,
    TO ellipticSurfaceD10S9,
    TO ellipticSurfaceD10S10,
    TO ellipticSurfaceD11S12,
    TO ellipticSurfaceD12S14L0,
    TO ellipticSurfaceD12S14Linfinite,
    TO ellipticSurfaceD12S13,
    TO specificEllipticSurfaceD13S16,
  },
}

```

```

    TO irregularEllipticSurfaceD12,
    TO specificEllipticAboSurfaceD12S13,
  },
  PARA {},
  SUBSECTION "6--secants and adjunction",
  UL {
    TO LeBarzN6,
    TO residualInQuintics,
    TO canonicalDivisor,
    TO selfIntersectionNumber,
  },
}

document {
  Key => adjunctionProcessData,
  Headline => "explains the output of the function adjunctionProcess",
  "We explain the output of the function adjunctionProcess from the package adjunctionForSurfaces.
  Notice the typo: adjointMatrix should be adjunctionProcess",
  help adjunctionProcess,
}

document {
  Key => symExtFunction,
  Headline => "symExt",
  "documentation of the symExt function from the BGG package",
  help symExt,
}

-- numerical functions

doc ///
Key
chiTable
(chiTable,ZZ,ZZ,ZZ)
Headline
compute a plausible cohomology tally for a smooth surface in P4
Usage
B = chiTable(d,sg,xO)
Inputs
d:ZZ
  a desired degree
sg:ZZ
  a desired sectional genus
xO:ZZ
  a desired Euler characteristic of O_X
Outputs
B:BettiTally
  plausible Betti tally of the cohomology of the desired ideal sheaf
Description
Text
  Since


$$\chi(l_X(m)) = \chi(O_{P^4}(m)) - \chi(O_X(m))$$


  one can compute  $\chi(l_X(m))$  from Riemann–Roch which depends only
  degree  $d$ , the sectional genus  $sg$ , the Euler characteristic  $xO$  and  $m$ .
  Assuming that  $l_X$  has natural cohomology for  $m$  in  $\{-4..8\}$  and  $m \rightarrow \chi(l_X(m))$  has enough
  sign changes, we get a plausible table.
Example
chiTable(11,10,1)
chiTable(12,13,1)
chiTable(12,13,2)

B=chiTable(12,13,3)
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=ellipticSurfaceD12S13 P4;
(degree X,sectionalGenus X)
B'=betti tateResolutionOfSurface(X,7)
B==B'
keyWithZeroValue=select(keys B,k->not member(k,keys B'))

```

```

B#(keyWithZeroValue_0)
Text
If  $\chi(l_X(m))$  in  $\mathbb{Z}[m]$  has an integral zero then B contains a superfluous key.
SeeAlso
tateResolutionOfSurface
ellipticSurfaceD12S13
sectionalGenus
chiO
///
///
chilTable(10,8,1)
tex chilTable(10,8,1)
chilTable(16,22,1)
apply(toList(-4..10),i->chil(i,16,24,1))
chilTable(13,14,1)
chilTable(13,15,1)
p=7
kk= $\mathbb{Z}/p$ 
E=kk[e_0..e_4,SkewCommutative=> true]
P4=kk[x_0..x_4]
while( --distinct values
  as=apply(5,i->random kk);
  #unique as != 5) do ()
m2x5=transpose matrix apply(5,i->{e_i,(i+1)*e_i})
elapsedTime tally apply(p^3,c->(
  m5x3=random(E^5,E^{3:-1});
  betti(hom=Hom(coker m2x5,coker m5x3,DegreeLimit=>4))))
///

doc ///
Key
sectionalGenus
(sectionalGenus, Ideal)
Headline
compute the sectional genus of a surface
Usage
sg = sectionalGenus X
Inputs
X: Ideal
ideal of a (smooth) projective surface
Outputs
sg:  $\mathbb{Z}$ 
the genus of a hyperplane section
Description
Text
The sectional genus of a projective surface is a part of the data provided by
the function genera.
Example
kk= $\mathbb{Z}/\text{nextPrime } 10^4$ 
P4=kk[x_0..x_4]
X=bordigaSurface P4;
sectionalGenus X
genera X
degree X == last genera X + 1
SeeAlso
genera
///

doc ///
Key
HdotK
(HdotK,  $\mathbb{Z}, \mathbb{Z}$ )
Headline
compute the intersection number H.K
Usage
HK = HdotK(d,sg)
Inputs
d:  $\mathbb{Z}$ 
degree of a smooth projective surface
Outputs
sg:  $\mathbb{Z}$ 

```

the [genus](#) of a hyperplane section

Description

Text

Let H denote the hyperplane [class](#) and K the canonical divisor [class](#) on a smooth projective surface.
By the adjunction formula

$$2sg - 2 = H \cdot (H + K)$$

one can compute $H \cdot K$ from the [degree](#) $d = H \cdot H$ and the sectional [genus](#) sg .

Example

```
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=bordigaSurface P4;
d=degree X
sg=sectionalGenus X
HdotK(d,sg)
```

SeeAlso

[sectionalGenus](#)

///

[doc](#) ///

Key

Ksquare

(Ksquare,ZZ,ZZ,ZZ)

Headline

compute the self [intersection number](#) K^2 of a smooth surface X in P^4 .

Usage

$k2 = \text{Ksquare}(d, sg, xO)$

Inputs

d :ZZ
the [degree](#) of X
 sg :ZZ
the sectional [genus](#) of X
 xO :ZZ
the Euler characteristic $(1 - q + pg)$ of O_X

Outputs

$k2$:ZZ

Description

Text

The self-[intersection number](#) of the canonical divisor of a smooth surface in P^4 is determined by the [degree](#) d , the sectional [genus](#) and the Euler characteristic of $\chi(O_X) = h^0(O_X) - h^1(O_X) + h^2(O_X)$:

$$d^2 - 10d - 5HK + 2K^2 + 12\chi(O_X) = 0$$

In general for a surface in P^5 the [right](#) hand side in this formula describes the [number](#) of non Cohen-Macaulay double points of the [image](#) under a projection from a point p in $P^5 \setminus \text{setminus } X$. Hence the name.

Example

```
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=bordigaSurface P4;
d=degree X
sg=sectionalGenus X
HK=HdotK(d,sg)
xO=1+(genera X)_0
xO==chiO(X)
K2=Ksquare(d,sg,xO)
d^2-10*d-5*HK-2*K2+12*xO==0
```

Text

The [complete intersection](#) of a quadric and a cubic in P^4 is a minimal $K3$ surface.

Example

```
X=ideal random(P4^1,P4^{-2,-3});
d=degree X
sg=sectionalGenus X
HK=HdotK(d,sg)
xO=1+(genera X)_0
xO==chiO(X)
```

```

K2=Ksquare(d,sg,xO)
d^2-10*d-5*HK-2*K2+12*xO==0
References
Hartshorne, Appendix A
SeeAlso
HdotK
sectionalGenus
chiO
irregularity
geometricGenus
///

doc ///
Key
chiO
(chiO,Ideal)
Headline
compute the Euler characteristic chiO(X)
Usage
xO = chiO(X)
Inputs
X:Ideal
  of a smooth projective variety
Outputs
xO:ZZ
  the Euler characteristic of the structure sheaf
Description
Text
The Euler characteristic of the structure sheaf  $\mathcal{O}_X$  is


$$\chi(\mathcal{O}_X) = \sum_{i=0}^{\dim X} (-1)^i h^i(\mathcal{O}_X).$$


This quantity coincides with  $1 + (\text{genera } X)_0$  by Hirzebruch–Riemann–Roch.
Example
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=bordigaSurface P4;
xO=chiO(X)
xO=1+(genera X)_0
q=irregularity X
pg=geometricGenus X
1-q+pg==chiO(X)
References
Hartshorne, V
SeeAlso
irregularity
geometricGenus
///

doc ///
Key
irregularity
(irregularity,Ideal)
Headline
compute  $h^1(\mathcal{O}_X)$ 
Usage
q = irregularity X
Inputs
X:Ideal
  of a smooth projective surface
Outputs
q:ZZ
  the dimension of the cohomology group  $H^1(\mathcal{O}_X)$ 
Description
Text
Using sheaf cohomology we can compute this quantity for a smooth projective surface.
Example
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=irregularEllipticSurfaceD12 P4;
xO=chiO(X)

```

```

xO=1+(genera X)_0
q=irregularity X
pg=geometricGenus X
1-q+pg==chiO(X)
References
Hartshorne, V
SeeAlso
chiO
geometricGenus
///

doc ///
Key
geometricGenus
(geometricGenus,Ideal)
Headline
compute  $h^2(O_X)$ 
Usage
pg = geometricGenus
Inputs
X:Ideal
of a smooth projective surface
Outputs
pg:ZZ
the dimension of the cohomology group  $H^2(O_X)$ 
Description
Text
Using sheaf cohomology we can compute this quantity for a smooth projective surface.
Example
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=irregularEllipticSurfaceD12 P4;
pg=geometricGenus X
xO=chiO(X)
xO=1+(genera X)_0
q=irregularity X
1-q+pg==chiO(X)
References
Hartshorne, V
SeeAlso
chiO
irregularity
///

doc ///
Key
LeBarzN6
(LeBarzN6,ZZ,ZZ,ZZ)
Headline
compute the value of Le Barz formula for 6-secants lines
Usage
N6 = LeBarzN6(d,sg,xO)
Inputs
d:ZZ
degree of a smooth projective surface in  $P^4$ 
sg:ZZ
the sectional genus of  $X$ 
oX:ZZ
the Euler characteristic of the structure sheaf
Outputs
N6:ZZ
the expected number of 6-secant line plus the number  $(-1)$  lines.
Description
Text
If there only finitely many 6-secant lines then LeBarz formula computes
the sum of the number of 6-secants lines (with multiplicities) and the number  $(-1)$ 
lines.
Example
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]

```

```

X=enriquesSurfaceOfDegree9 P4;
d=degree X
sg=sectionalGenus X
xO=1+(genera X)_0
N6=LeBarzN6(d,sg,xO)
minimalBetti X
K2=Ksquare(d,sg,xO)
HK=HdotK(d,sg)
d+2*HK+K2==10
d+3*HK+2*K2==10

```

Text

Since the homogeneous ideal of this surface is generated by quintics, there are no 6-secant lines by Bezout. Thus there is a (-1) line, which will be blown down by the first adjunction map. The image of the adjunction map defined by $|H+K|$ is a surface X_1 in $P^{\wedge\{sg-1\}}$ of degree $(H+K)^2=10$, $K_1^2=K_2+1=0$ and sectional genus $sg_1=6$ since $2sg_1-2=(H+K).(H+2K)=10$. It follows that X_1 is a minimal Enriques surface.

On the other hand the surface

Example

```

X=nonspecialAlexanderSurface P4;
d=degree X
sg=sectionalGenus X
xO=1+(genera X)_0
minimalBetti X

```

Text

has the same numerical invariants and visibly a 6-secant line.

Example

```

L=residualInQuintics X
dim L, degree L
dim (X+L),degree (X+L)

```

References

\textit{P. Le Barz}, Formules pour les multiseccants des surfaces, C. R. Acad. Sci., Paris, Sér. I 292, 797– 800 (1981) Zbl 0492.14045)

SeeAlso

```

residualInQuintics
///

```

doc ///

Key

```

residualInQuintics
(residualInQuintics,Ideal)

```

Headline

compute the residual scheme of X in the ideal generated by the quintics

Usage

```

Z = residualInQuintics X

```

Inputs

```

X:Ideal
homogenous ideal of a smooth projective surface in P4

```

Outputs

```

Z:Ideal
the residual ideal in the ideal generated by the quintics in X

```

Description

Text

If an ideal of a surface in P^4 has 6-secant lines, then any such line is contained in vanishing loci of the residual ideal Z by Bezout. This allows us frequently to compute the number of 6-secant lines and hence the number of (-1) lines by LeBarz 6-secant formula.

Example

```

kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=vBELSurface P4;
d=degree X
sg=sectionalGenus X
xO=1+(genera X)_0
N6=LeBarzN6(d,sg,xO)
minimalBetti X
Z=residualInQuintics X;
dim Z, degree Z
tally apply(decompose Z,L->(dim L, degree L, degree (L+X)))

```

Text

Thus there are two 6-secant [lines](#) and by Le Barz 6-secant formula, the [first adjunction map](#) defined by $|H+K|$ contracts five (-1) [lines](#).

Example

```
K2=Ksquare(d,sg,xO)
"elapsedTime (numList,L2,L3,J)=adjunctionProcess(X,2)";
"numList=={(4, 11, 11), 5, (10, 18, 9), 14, (8, 8, 1)}";
"degree J == 9-1";
"sectionalGenus J==1";
K2==8-5-14-1
```

Text

Thus this surface arises as a blow-up of a Del Pezzo surface of [degree 8](#) in $14+5$ points.

References

[\textit{P. Le Barz}, Formules pour les multiseccants des surfaces, C. R. Acad. Sci., Paris, Sér. I 292, 797– 800 \(1981\) Zbl 0492.14045\)](#)

SeeAlso

LeBarzN6
sectionalGenus
Ksquare

///

doc ///

Key

canonicalDivisor
(canonicalDivisor,[Ideal](#))

Headline

compute a canonical divisor [on](#) a surface with $pg>0$

Usage

D = canonicalDivisor X

Inputs

X:[Ideal](#)

homogenous [ideal](#) of a smooth projective surface in P^4

Outputs

D:[Ideal](#)

the [ideal](#) of an effective canonical divisor [on](#) X

Description

Text

If X is a smooth projective surface with $pg>0$, then X has an effective canonical divisor, which can be computed from the [presentation matrix](#) $\omega_X = \text{Ext}^1(X, P^4\{-5\})$.

Example

```
kk=ZZ/nextPrime 10^4;
P4=kk[x_0..x_4];
X=K3surfaceD8 P4;
pg=geometricGenus X
d=degree X
sg=sectionalGenus X
xO=chiO(X)
D=canonicalDivisor X;
degree D==2
genus D==0
selfIntersectionNumber(X,D)==-1
sectionalGenus X == 6
Ksquare(d,sg,xO)
```

Text

Thus X is the projection from the tangent plane at a point p in a minimal K3 surface X2 in P^7 of [degree 12](#) $=8+4$.
(Check this again!)

References

SeeAlso

chiO
sectionalGenus
canonicalDivisor
Ksquare

///

doc ///

Key

selfIntersectionNumber

(selfIntersectionNumber,Ideal,Ideal)

Headline

compute the self intersection number of an effective divisor on a smooth surface

Usage

DdotD = selfIntersectionNumber(X,D)

Inputs

X:Ideal

homogenous ideal of a smooth projective surface

D:Ideal

homogeneous ideal of an effective divisor D on X

Outputs

DdotD:ZZ

the self intersection number of D on X

Description

Text

The self intersection number of effective divisor can be compute from the genus g_1 of D and the genus g_2 of $2D$, since

$$2g_1 - 2 = D \cdot (D + K) \text{ and } 2g_2 - 2 = 2D \cdot (2D + K)$$

holds.

Example

```
kk=ZZ/nextPrime 10^4;
```

```
P4=kk[x_0..x_4];
```

```
X=K3surfaceD8 P4;
```

```
pg=geometricGenus X
```

```
d=degree X
```

```
sg=sectionalGenus X
```

```
xO=chiO(X)
```

```
D=canonicalDivisor X;
```

```
degree D==2
```

```
genus D==0
```

```
selfIntersectionNumber(X,D)==-1
```

```
sectionalGenus X == 6
```

```
Ksquare(d,sg,xO)
```

Text

Thus X is the projection from the tangent plane at a point p of a minimal K3 surface X_2 in P^7 of degree $12=8+4$.

References

SeeAlso

chiO

sectionalGenus

canonicalDivisor

Ksquare

///

doc ///

Key

tateResolutionOfSurface

(tateResolutionOfSurface,Ideal)

(tateResolutionOfSurface,Ideal,ZZ)

Headline

compute the Tate resolution of the ideal sheaf of a surface in P^4

Usage

T = tateResolutionOfSurface X

T = tateResolutionOfSurface(X,n)

Inputs

X:Ideal

homogenous ideal of a smooth projective surface in P^4

Outputs

T: ChainComplex

the Tate resolution of the ideal sheaf of surface in P^4

Description

Text

The Tate resolution T of a coherent sheaf F on a projective space P^n is an infinite exact complex of free modules over an exterior algebra E encoding the cohomology groups of F :

$T^d(F) = \sum_{i=0}^n \text{Hom}_{kk}(E, H^i(P_n, F(d-i)))$.

For details see [EFS].

In case of a surface the interesting cohomology groups lie in the range $d = -1..7$.
From the betti table of T we can read of the dimension of certain cohomology group.

Example

```
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=ellipticSurfaceD12S14Linfinite P4;
minimalBetti X
T=tateResolutionOfSurface X;
"elapsedTime geometricGenus X == 2"; -- 52.2s elapsed
betti T
33=5*8-9+2
```

Text

For example, the entry 1 in homological degree -1 is $h^4(I_X(-5)) = h^4(O_{P4}(-5)) = 1$.
The entry 2 in homological degree 3 is $h^3(I_X) = h^2(O_X) = pg$.
 $h^0(I_X(6)) = 33 = 5*8 - 9 + 2$ which we can read of betti numbers of the minimal free resolution of X.

Note that it might be faster to compute the geometric genus pg and the irregularity q of a surface by using the Tate resolution instead of sheaf cohomology.

If the homogeneous ideal is generated by forms of degree ≤ 6 , then the truncation used in the computation can be chosen to be 6. If there are larger degree generators we might need a larger truncation.

Example

```
X=irregularEllipticSurfaceD12 P4;
minimalBetti X
betti tateResolutionOfSurface(X,7)
```

Text

So the irregularity of this surface is $q=1$ and the geometricGenus is $pg=3$.
It is a minimal elliptic surface.

Example

```
sg=sectionalGenus X
d=degree X
xO=chiO(X)
Ksquare(d,sg,xO)==0
HdotK(d,sg)
LeBarzN6(d,sg,xO)
residual=residualInQuintics X;
dim residual,degree residual
primaryDecomposition residual
cH=primaryDecomposition saturate (X+ideal x_2)
tally apply(cH,c->(dim c, degree c, degree radical c, minimalBetti c))
```

Text

Say something sensible.

References

\textit{D. Eisenbud, G. Floystad, F.-O. Schreyer} Sheaf cohomology and free resolutions over exterior algebras, Trans. Am. Math. Soc. 355, No. 11, 4397-4426 (2003; Zbl 1063.14021)

SeeAlso

```
geometricGenus
irregularity
chiO
```

///

doc ///

Key

```
tangentToMonad
(tangentToMonad,Ideal)
```

Headline

Compute the tangent space to the monad for a surface

Usage

```
m = tangentToMonad X
```

Inputs

```
X:Ideal
homogenous ideal of a smooth projective surface in P4
```

Outputs

```
m:Matrix
which describes the first order deformation space for the monad of X
```

Description

Text

add some explanation

Example

```
kk=ZZ/19
P4=kk[x_0..x_4]
E=kk[e_0..e_4,SkewCommutative=>true]
minimalBetti(X=specificAboSurface(P4,E,1))
m=tangentToMonad X;
rank source m
```

SeeAlso

///

///

Example

```
sg=sectionalGenus X
d=degree X
xO=chiO(X)
Ksquare(d,sg,xO)==0
HdotK(d,sg)
LeBarzN6(d,sg,xO)
residual=residualInQuintics X;
dim residual,degree residual
primaryDecomposition residual
cH=primaryDecomposition saturate (X+ideal x_2)
tally apply(cH,c->(dim c, degree c, degree radical c, minimalBetti c))
```

Text

Say something sensible.

///

—* schreyer surfaces *

doc ///

Key

moduleFromSchreyerSurface
(moduleFromSchreyerSurface, Ideal)

Headline

compute the H^1 -module of the ideal sheaf of X

Usage

M = moduleFromSchreyerSurface X

Inputs

X:Ideal
of a Schreyer surface in P4

Outputs

M:Ideal
the ideal defining the H^1 -module of the ideal sheaf of X

Description

Text

The H^1 -module of a Schreyer surface is a finite length module with Hilbert function (1,5,5) with at least two extra syzygies.

Example

```
P4=ZZ/3[x_0..x_4];
(Ms,Types)=exampleOfSchreyerSurfaces P4;
tally apply(Ms,M->minimalBetti M)
X=schreyerSurfaceFromModule Ms_1;
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
```

///

doc ///

Key

schreyerSurfaceFromModule
(schreyerSurfaceFromModule, Ideal)

Headline

compute a Schreyer Surface with given the H^1 -module of the ideal sheaf

Usage

X = schreyerSurfaceFromModule M

Inputs

M: **ideal**
the **ideal** defining the H^1 -**module** of the **ideal sheaf** of X

Outputs

X: **ideal**
the **ideal** of a Schreyer surface

Description

Text

The H^1 -**module** of a Schreyer surface is a finite **length module** with **Hilbert** function (1,5,5) with at least two extra syzygies.

Example

```
P4=ZZ/3[x_0..x_4];
(Ms,Types)=exampleOfSchreyerSurfaces P4;
tally apply(Ms,M->minimalBetti M)
X=schreyerSurfaceFromModule Ms_1;
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
///
```

doc ///

Key

schreyerSurfaceWith2LinearSyzygies
(schreyerSurfaceWith2LinearSyzygies, **Ring**)
[schreyerSurfaceWith2LinearSyzygies,Smooth]

Headline

compute a rational Schreyer surface whose H^1 -**module** has 4 extra syzygies

Usage

X = schreyerSurfaceWith2LinearSyzygies(P4)

Inputs

P4: **Ring**
the coordinate **ring** of P4

Outputs

X: **ideal**
the **ideal** of a smooth Schreyer surface

Description

Text

The construction is a 2 **step** liaison construction.

The desired surface has a residual scheme $R=X^5:X$ consisting of the **union** of 3 planes.

A general (5,5) **complete intersection** ci has as residual scheme $ci:X=R \cup Y$ with Y a surface of **degree** 11 which lies on two quartics. The (4,4) **complete intersection** $ci2$ has residual $Z=ci2:Y$ of **degree** 5 which decomposes in a cubic scroll and a quadric surface which **intersect** along the directrix of the scroll and two non-CM points of Z .

Example

```
kk=ZZ/nextPrime(2*10^3);P4=kk[x_0..x_4];
X=schreyerSurfaceWith2LinearSyzygies(P4);
elapsedTime X=schreyerSurfaceWith2LinearSyzygies(P4);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
X5=ideal (gens X){0..4};
R=X5:X;
minimalBetti radical R
tally apply(decompose R,c->(dim c, degree c, minimalBetti c))
ci=ideal( gens X*random(source gens X,P4^{2:-5}));
Y=(ci:X):R;
degree Y,betti(fY=res Y)
nCM=decompose ann coker transpose fY.dd_3
ci2=ideal (gens Y){0,1};
Z=ci2:Y;
minimalBetti Z
cZ=decompose Z;
tally apply(cZ,c->(dim c, degree c, minimalBetti c))
```

Text

The construction is a reversal of this linkage. Note that both Y and Z are **not** Cohen-Macaulay in two (common) points.

Example

```
intersectionOftheTwoComponentsOfZ=sum(cZ);
apply(cl=decompose intersectionOftheTwoComponentsOfZ,c->(dim c, degree c))
```

```

cl, cl_{1,2}==nCM
planes=decompose R
matrix apply(planes,p2->apply(nCM,p->dim(p2+p)))
matrix apply(planes,p2->apply(planes,p2'->dim(p2+p2')))
dim(radical R+Z),degree(radical R+Z)
matrix apply(planes,p2->apply(cZ,c->degree(p2+c)))
m3x2=(res cZ_1).dd_2
syz transpose (m3x2%cl_0) -- => cl_0 is the directrix of the scroll
///

doc ///
Key
schreyerSurfaceWith2or3LinearSyzygies
(schreyerSurfaceWith2or3LinearSyzygies, Ring, ZZ)
[schreyerSurfaceWith2or3LinearSyzygies, Smooth]
Headline
compute a rational Schreyer surface whose  $H^1$ -module has 4 or 5 extra syzygies
Usage
X = schreyerSurfaceWith2or3LinearSyzygies(P4,s)
Inputs
P4: Ring
the coordinate ring of P4
s: ZZ
a number s=2 or s=3
Outputs
X: Ideal
the ideal of a smooth Schreyer surface
Description
Text
The construction is a 2 step liaison construction.
In case of s=2, the desired surface has a residual scheme  $R=X^5:X$  consisting of the union of 3 planes.
A general (5,5) complete intersection ci has as residual scheme  $ci:X=R \cup Y$  with
Y a surface of degree 11 which lies on two quartics. The (4,4) complete intersection
ci2 has residual  $Z=ci^2:Y$  of degree 5 which decomposes in a cubic scroll and a quadric surface
which intersect along the directrix of the scroll and two non-CM points of Z.
Example
kk=ZZ/nextPrime(2*10^3);P4=kk[x_0..x_4];
elapsedTime X=schreyerSurfaceWith2or3LinearSyzygies(P4,2);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
R=residualInQuintics X;
tally apply(primaryDecomposition R,c->(dim c, degree c, minimalBetti c))
ci=ideal( gens X*random(source gens X,P4^{2:-5}));
Y=(ci:X):R;
degree (ci:X)
degree Y,betti(fY=res Y)
nCM=decompose ann coker transpose fY.dd_3
ci2=ideal (gens Y){0,1};
Z=ci2:Y;
minimalBetti Z
cZ=decompose Z;
tally apply(cZ,c->(dim c, degree c, minimalBetti c))
Text
The construction is a reversal of this linkage. Note that both Y and Z are not Cohen-Macaulay
in two (common) points.
Example
intersectionOftheTwoComponentsOfZ=sum(cZ);
apply(cl=decompose intersectionOftheTwoComponentsOfZ,c->(dim c, degree c))
cl, cl_{1,2}==nCM
planes=decompose R
matrix apply(planes,p2->apply(nCM,p->dim(p2+p)))
matrix apply(planes,p2->apply(planes,p2'->dim(p2+p2')))
dim(radical R+Z),degree(radical R+Z)
matrix apply(planes,p2->apply(cZ,c->degree(p2+c)))
m3x2=(res cZ_1).dd_2
cl_0
syz transpose (m3x2%cl_0) -- => cl_0 is the directrix of the scroll
Text
In case s=3 the residualInQuintics scheme consists of 5 planes.
Example

```

```

elapsedTime X=schreyerSurfaceWith2or3LinearSyzygies(P4,3);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
R=residualInQuintics X;
cR=decompose R;
tally apply(cR,c->(dim c, degree c))
ci=ideal( gens X*random(source gens X,P4^{2:-5}));
Y=(ci:X):R;
degree (ci:X)
degree Y,betti(fY=res Y), sectionalGenus Y
apply(cR,p->dim trim(p+Y))
matrix apply(cR[{Y}],p->apply(cR[{Y}],q-> dim(p+q)))
beti tateResolutionOfSurface Y
beti tateResolutionOfSurface X
Ksquare(9,9,4)
HdotK(9,9)
K=canonicalDivisor Y;
degree Y, degree K
saturate ideal singularLocus Y
selfIntersectionNumber(Y,K)
///

doc ///
Key
unirationalConstructionOfSchreyerSurface
(unirationalConstructionOfSchreyerSurface, Ring)
Headline
compute a rational Schreyer surface whose  $H^1$ -module has 5 extra syzygies
Usage
X = unirationalConstructionOfSchreyerSurface P4
Inputs
P4:Ring
the coordinate ring of P4
Outputs
X:Ideal
the ideal of a smooth Schreyer surface
Description
Text
The desired surface has a residual scheme  $R=X^5:X$  consisting of the union of 5 planes.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=unirationalConstructionOfSchreyerSurface(P4);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
X5=ideal (gens X){0..4};
R=X5:X;
minimalBetti R
planes=decompose R
tangentDimension M
tally apply(planes,p->tally apply(decompose(p+X),c->(dim c, degree c, betti c)))
sixSecants1=apply(planes,p-> ideal (gens intersect drop(select(decompose(p+X),c->dim c==1),1))_{0,1,2});
sixSecants2=apply(5,i->trim (planes_i+planes_{(i+1)%5}));
sixSecants=sixSecants1|sixSecants2
tally apply(sixSecants, l-> (beti l,dim l, degree (l+X)))
LeBarzN6(11,10,1)=10
Text
Each of the five planes intersects X in a plane quartic curve and three points.
The sixSecants are the five intersection lines of the planes and the five lines spanned by two of
the special points in each plane.
///

///
LeBarzN6(11,10,1)
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
planes:=apply(5,i->ideal(P4_i,P4_{(i+1)%5}));

```

```

ps := intersect planes;
Ls:=apply(planes,i->i+ideal random(1,P4)); --10 param = P2^5
L:= intersect Ls;minimalBetti L, dim L, degree L, genus L
cub:=ideal(gens L*random(source gens L,P4^{3}));-- P9 of choices for cub
C:=(intersect planes+cub):L;
minimalBetti C,dim C, degree C, genus C
netList (cC=decompose C)
beti(fC:=res C)
M:=ideal fC.dd_4^{1..10};
cub1:=ideal(gens L*random(source gens L,P4^{3}));-- P9 of choices for cub
C1:=(intersect planes+cub1):L;
minimalBetti (intersect planes+cub)
beti(fC1:=res C1)
M1:=ideal fC1.dd_4^{1..10};
M==M1 -- => M does not depend on the choice of cub
beti (A=saturate ann(coker fC.dd_2^{6..10})_{5..19}))
minimalBetti A
minimalBetti coker fC.dd_2^{6..10})_{5..19} -- direct sum of 5 lines
tangentDimension M, 10+25-5
X=schreyerSurfaceFromModule M; 10+24-4+2*3
minimalBetti X
X5=ideal (gens X)_{0..4};
R=X5:X;
minimalBetti R
cR=decompose R
tally apply(cR,p->tally apply(decompose(p+X),c->(dim c, degree c)))
netList cC
netList apply(planes,p->(p,select(decompose(p+X),c->dim c==1)))
netList (sixSecants1=apply(planes,p-> ideal (gens intersect drop(select(decompose(p+X),c->dim c==1))_{0,1,2})))
netList unique (apply(Ls,L->trim L)|decompose A)
netList (sixSecants2=apply(planes,p-> ideal (gens intersect(select(decompose(p+X),c->dim c==1))_{0,1})_{0,1,2})))
sixSecants3=apply(5,i->trim (planes_i+planes_{(i+1)%5}))
sixSecants=sixSecants1|sixSecants2
tally apply(sixSecants, l-> (dim l, degree (l+X)))
///

```

doc ///

Key

specialEnriquesSchreyerSurface

(specialEnriquesSchreyerSurface, Ring)

Headline

compute a Enriques Schreyer surface whose H^1 -module has 5 extra syzygies

Usage

X = specialEnriquesSchreyerSurface P4

Inputs

P4:Ring

the coordinate ring of P4

Outputs

X:Ideal

the ideal of a smooth Schreyer surface

Description

Text

The desired surface has a residual scheme $R=X_5:X$ which is a quintic elliptic scroll.

The H^1 -module is defined by the sum of the ideals of two elliptic curves on the scroll.

Thus the construction needs a point p on the Bring curve and the two points on the conic of elliptic normal curves of degree 5. Over a finite field such data are easily found by a random search, whose running time is independent of the finite ground field

The two points on the conic are the intersection of the conic with the polar line to the point p of the conic, [Hulek,199x].

The rest of the construction is unirational.

Example

kk=ZZ/nextPrime 10^3;

P4=kk[x_0..x_4];

X=specialEnriquesSchreyerSurface(P4);

minimalBetti X

M=moduleFromSchreyerSurface X;

minimalBetti M

X5=ideal (gens X)_{0..4};

R=X5:X;

minimalBetti R

```

tangentDimension M==25
Text
=> these surfaces do not form a complete family
///

///
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4]
elapsedTime X=specialEnriquesSchreyerSurface(P4);
X5=ideal (gens X)_{0..4};
R=X5:X;
minimalBetti R
M=moduleFromSchreyerSurface X;minimalBetti M
fM=res M
m5x15=(transpose fM.dd_3_{0..4}^{0..14})*P4^{-5};
fm5x15=res coker m5x15
betti fm5x15
isIsomorphic(coker m5x15,coker transpose fm5x15.dd_3)
-- => m5x15 is the presentation matrix of the module of global sections of a 2-torsion bundle in E3
E3=ann coker m5x15;
minimalBetti E3
minimalBetti E3

E3+R==E3
cub=ideal(gens E3*random(source gens E3,P4^{-3}));
betti(A=R+cub:E3); dim A, degree A, genus A
betti saturate A
singA=A+minors(3,jacobian A);
saturate singA

elapsedTime cA=decompose A;
apply(decompose(R+quad),c->(dim c, degree c, genus c))
elapsedTime X=specialEnriquesSchreyerSurface(P4);
betti(fX=res(X,LengthLimit=>2))
elapsedTime betti(N=Hom(module X,P4^1/X)) -- 362.604 seconds elapsed

--phis={homomorphism (N_{4})};
phis=apply(toList(5..16),i->homomorphism N_{i});
As=apply(phis,phi->matrix{apply(15,i->lift(phi_(0,i),P4))});
Bs1=apply(As,A->A*fX.dd_2/fX.dd_1);
betti Bs_1
Bs=apply(Bs1,B->B^{5..14}_{0..2})
T=kk[t_0..t_{#Bs-1}]
m10x3=sum(#Bs,i->t_i*sub(Bs_i,kk))
betti(J2=minors(2,m10x3))
betti(saturate J2)
betti(J=minors(3,m10x3))
codim J == 10-3+1
dim J

P4xT=P4**T
betti(A=sum(#As,i->P4xT_{(5+i)*sub(As_i,P4xT)}))
betti(B=sum(#Bs1,i->P4xT_{(5+i)*sub(Bs1_i,P4xT)}))
betti(I=ideal(A*B))
b6=sub(basis(6,P4),P4xT)
betti (I1 =trim ideal sub(contract(b6, gens I),T))
b7=sub(basis(7,P4),P4xT)
betti (I2 =trim ideal sub(contract(b7, gens I),T))
betti trim (I1+I2)
dim I1
betti basis(2,T)
binomial(13,2)
///

doc ///
Key
specificSchreyerSurface
(specificSchreyerSurface, Ring, Number)
Headline
compute a smooth Schreyer surface with given H^1-module
Usage

```

X = specificSchreyerSurface(P4,k)

Inputs

P4: Ideal

coordinate ring of P4 over a ground field of characteristic 3

k: Number

a number between 0 and 9 specifying the specific H^1 -module to use.

Outputs

X: Ideal

ideal of a Schreyer surface

Description

Text

The function returns one of ten specific smooth schreyer surfaces.

It prints the corresponding adjunction process data.

The corresponding H^1 -module are precomputed and stored in the function exampleOfSchreyerSurfaces.

Example

```
P4=ZZ/3[x_0..x_4];
```

```
X=specificSchreyerSurface(P4,1);
```

```
(d,sg)=(degree X, sectionalGenus X)
```

```
Ksquare(d,sg,1)=-6
```

```
LeBarzN6(d,sg,1)=10
```

```
minimalBetti X
```

```
betti(X5=ideal (gens X)_{0..4})
```

```
betti(residual=X5:X)
```

```
dim residual, degree residual
```

```
tally apply(primaryDecomposition residual,c-> (
  (dim c, degree c, degree (c+X))))
```

Text

There are 6 six-secant lines grouped in Frobenius orbis.

So there should be 4 (-1) lines. Indeed, the adjunction data say this.

The last surface in the adjunction process is a conic bundle with

$6+8-5=9$ singular fibers.

The construction of X uses a special Hartshorne-Rao module M.

Example

```
betti tateResolutionOfSurface X
```

```
M=moduleFromSchreyerSurface X;
```

```
minimalBetti M
```

```
tangentDimension M==36
```

Text

Thus the space of good modules in the Grassmannian $G(5,15)$ of dimension 50 is smooth of the expected codimension 14 at our point M.

Hence X lifts to a surface defined over some number field, which gives a surface over

CC.

SeeAlso

adjunctionProcessData

LeBarzN6

Ksquare

tateResolutionOfSurface

moduleFromSchreyerSurface

tangentDimension

exampleOfSchreyerSurfaces

///

doc ///

Key

exampleOfSchreyerSurfaces

(exampleOfSchreyerSurfaces, Ring)

Headline

read the list of precomputed H^1 -modules of Schreyer surfaces

Usage

(Ms,types)= exampleOfSchreyerSurfaces P4

Inputs

P4: Ideal

coordinate ring of P4 over a ground field of characteristic 3

Outputs

Ms: List

of H^1 -modules of Schreyer surfaces

types: List

of precomputed adjunction type data of the corresponding surfaces

Description

Text

The function reads lists of precomputed H^1 -modules and adjunction types

Example

```
P4=ZZ/3[x_0..x_4];
(Ms,types)=exampleOfSchreyerSurfaces P4;
tally apply(Ms,M->minimalBetti M)
netList apply(#Ms,i->(minimalBetti Ms_i, types_i))
///
///
-- additional information
P4=ZZ/3[x_0..x_4];
(Ms,types)=exampleOfSchreyerSurfaces P4;
tally apply(Ms,M->minimalBetti M)
#Ms
netList apply(#Ms,i->(minimalBetti Ms_i, types_i))
elapsedTime Xs=apply(9,i->schreyerSurfaceFromModule Ms_i); -- 665.508s elapsed
#Xs
netList apply(9,i->(X=Xs_i;
  R=residualInQuintics X;
  type=types_i;
  s=#type;
  curves=reverse flatten apply(sub((s-1)/2,ZZ),j->type_{2*j+1});
  lastSurface=type_(s-1);
  residual=tally (apply(decompose R,c->(dim c-1, degree c,
    (dim (c+X)-1,degree(c+X)))));
  (curves,lastSurface,residual))
)
--
+-----+
o46 = |({0, 0, 1, 5}, (10, 20, 11), Tally{(1, 1, (0, 6)) => 1 })|
|      (1, 3, (0, 18)) => 1 |
+-----+
|({0, 0, 1, 4}, (7, 11, 5), Tally{(1, 1, (0, 6)) => 2 }) |
|      (1, 2, (0, 12)) => 1 |
+-----+
|({4, 2, 1, 3}, (4, 4, 1), Tally{(1, 7, (0, 42)) => 1}) |
+-----+
|({3, 0, 2, 3}, (5, 6, 2), Tally{(1, 1, (0, 6)) => 2}) |
|      (2, 1, (1, 4)) => 1 |
+-----+
|({2, 2, 3, 2}, (3, 3, 1), Tally{(1, 2, (0, 12)) => 1}) |
|      (2, 2, (1, 8)) => 1 |
+-----+
|({8, 1, 3, 2}, (3, 2, 0), Tally{(1, 1, (0, 6)) => 1}) |
|      (2, 2, (1, 8)) => 1 |
+-----+
|({7, 4, 2, 2}, (2, 1, 0), Tally{(1, 2, (0, 12)) => 1}) |
|      (1, 3, (0, 18)) => 1 |
|      (2, 1, (1, 4)) => 1 |
+-----+
|({4, 4, 1}, (6, 7, 2), Tally{(1, 1, (0, 6)) => 1}) |
|      (2, 1, (1, 4)) => 3 |
+-----+
|({5, 6, 0}, (5, 5, 1), Tally{(2, 5, (1, 20)) => 1}) |
+-----+
*_
-- compare with the table for schreyer surfaces
///

doc ///
Key
schreyerSurface
(schreyerSurface, Ring, Number)
[schreyerSurface,Smooth]
[schreyerSurface,Verbose]
Headline
find a random Schreyer surface
Usage
X = schreyerSurface(P4,s)
```

Inputs

P4: Ideal

coordinate ring of P4 over a ground field of characteristic 3

s: Number

the number of desired extra syzygies

Outputs

X: Ideal

the ideal of a Schreyer surface

Description

Text

It searches for a suitable H^1 -module with Hilbert function (1,5,5) and $s > 1$ extra syzygies by searching in the codimension $6+2(s-2)$ subspace of modules with one extra syzygy, and computes the corresponding surface. To find an example one has to check about 3^6 examples of modules.

Example

```
P4=ZZ/3[x_0..x_4];
setRandomSeed("find one fairly fast");
elapsedTime X=schreyerSurface(P4,2,Smooth=>false,Verbose=>true);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
```

```
setRandomSeed("also fairly fast");
elapsedTime X=schreyerSurface(P4,3,Smooth=>false);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
```

SeeAlso

findRandomSmoothSchreyerSurface

///

doc ///

Key

findRandomSchreyerSurface

(findRandomSchreyerSurface, Ring)

(findRandomSchreyerSurface, Ring, Number)

Headline

find a random Schreyer surface

Usage

X = findRandomSchreyerSurface P4

X = findRandomSchreyerSurface(P4,s)

Inputs

P4: Ideal

coordinate ring of P4 over a ground field of characteristic 3

s: Number

the number of desired extra syzygies

Outputs

X: Ideal

the ideal of a possibly singular Schreyer surface

Description

Text

It searches for a suitable H^1 -module with Hilbert function (1,5,5) and two extra syzygies by searching in the codimension 8 subspace of modules with one extra syzygy, and computes the corresponding surface. To find an example one has to check about 3^8 examples of modules.

Example

```
P4=ZZ/3[x_0..x_4];
setRandomSeed("find one fairly fast");
elapsedTime X=findRandomSchreyerSurface P4;
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
```

```
setRandomSeed("also fairly fast");
elapsedTime X=findRandomSchreyerSurface(P4,3);
minimalBetti X
M=moduleFromSchreyerSurface X;
minimalBetti M
```

SeeAlso

findRandomSmoothSchreyerSurface

///

```

doc ///
Key
findRandomSmoothSchreyerSurface
(findRandomSmoothSchreyerSurface, Ring, Number)
[findRandomSmoothSchreyerSurface, Verbose]
Headline
find a random smooth Schreyer surface
Usage
X = findRandomSmoothSchreyerSurface P4
X = findRandomSmoothSchreyerSurface(P4,s)
Inputs
P4:Ring
the coordinate ring of P4 over a ground field of characteristic 3 (or other small prime fields)
s:Number
the number of desired extra syzygies
Outputs
X:Ideal
the ideal of a smooth Schreyer surface
Description
Text
It searches for a suitable  $H^1$ -module with Hilbert function (1,5,5) and  $\min(2,s)$  extra syzygies by searching in the
codimension 8 subspace of modules with one extra syzygy, and computes the corresponding surface
and checks it smoothness. Since many  $H^1$ -modules lead to singular surfaces one has to check
more than  $3^8$  examples of modules.
Example
P4=ZZ/3[x_0..x_4];
setRandomSeed("carefully chosen good randomSeed ");
elapsedTime X=findRandomSmoothSchreyerSurface(P4,2);
minimalBetti X
singX=X+minors(2,jacobian X);
dim saturate singX== -1
SeeAlso
findRandomSchreyerSurface
///

```

```

doc ///
Key
collectSchreyerSurfaces
(collectSchreyerSurfaces, List, List, Number, Number)
(collectSchreyerSurfaces, List, List, Number)
Headline
collect random smooth Schreyer surfaces
Usage
(adj',Ms')= collectSchreyerSurfaces(adjTypes,Ms,s,N)
Inputs
adjTypes:List
of already discovered adjunction types
Ms: List
of ideals leading to the desired  $H^1$ -modules
s:Number
the number of desired extra syzygies
N:Number
the desired number of new modules
Outputs
adj':List
of adjunction types
Ms' : List
of modules
Description
Text
It searches for a suitable  $H^1$ -module with Hilbert function (1,5,5) and  $\min(2,s)$  extra syzygies by searching in the
codimension 8 subspace of modules with one extra syzygy, and computes the corresponding surface
and checks it smoothness. Since many  $H^1$ -modules lead to singular surfaces one has to check
more than  $3^8$  examples of modules.
Example
P4=ZZ/3[x_0..x_4];
setRandomSeed("carefully chosen good randomSeed ");
(Ms,adjTypes)=exampleOfSchreyerSurfaces P4;
netList adjTypes
Text

```

The following command takes too much time

Example

```
#Ms
"elapsedTime (adj',Ms')=collectSchreyerSurfaces(adjTypes,Ms,2,1);"
```

SeeAlso

exampleOfSchreyerSurfaces
///

doc ///

Key

tangentDimension
(tangentDimension, Ideal)

Headline

compute the dimension of the tangent space of the strata with s extra syzygies at the point M

Usage

d=tangentDimension M

Inputs

M:Ideal
ideal defining a H^1 -module with Hilbert function (1,5,5) with s extra syzygies

Outputs

d:Number
dimension of the tangent space of the corresponding strata at the given point M

Description

Text

To prove the existence of a lift of the corresponding surface to characteristic 0, it suffices to prove that the tangent space has dimension $d=36-\dim G(2,s)=36-2*(s-2)$.

Example

```
P4=ZZ/3[x_0..x_4];
(Ms,types)=exampleOfSchreyerSurfaces P4;
--tally apply(Ms,M->minimalBetti M)
--tally apply(Ms, M->tangentDimension M)
elapsedTime netList apply(Ms,M->(minimalBetti M, tangentDimension M))
--elapsedTime Xs=apply(Ms,M->schreyerSurfaceFromModule M);
--tally apply(Xs,X -> (singX=X+minors(2,jacobian X); dim saturate sing X)
```

Text

This proves that the surfaces precomputed Via exampleOfSchreyerSurfaces all lift to smooth surfaces over some algebraic number field (of characteristic 0).

///

-* Abo-Ranestad surfaces *-

doc ///

Key

veroneselImagesInG25
(veroneselImagesInG25, Matrix)

Headline

compute the Veronese images of P2 and P3 in the Grassmannian G25 and their intersection

Usage

(pts,vP2,vP3,g25) = veroneselImagesInG25(m4x2)

Inputs

m4x2: Matrix
the 4x2 matrix of linear forms over the exterior algebra

Outputs

pts:Ideal
the intersection points of the two Veronese images
vP2:Ideal
the ideal of the veronese surface
vP3:Ideal
the ideal of the veronese 3-fold
g25:Ideal
the ideal of the Grassmannian G(2,5) in P9.

Description

Text

It the Tate resolution of an Abo-Ranestad surface there are a 4x2 matrix m4x2 and a 2x3 matrix m2x3 with linear entries over the exterior algebra. These matrices define rational maps $P3 \rightarrow G(2,5)$ and $P2 \rightarrow G(2,5)$ and the type of the surface depends on how these images intersect in the Grassmannian G(2,5). It turns out that the number of

(-1) lines on the surface will coincide with the number of intersection points of the images + 1.
This function verifies this assertion in an example.

Example

```
kk=ZZ/nextPrime 10^3; P4:=kk[x_0..x_4];
n=7;
elapsedTime (X,m4x2) = aboRanestadSurface(P4,n,Special=>2);
(pts,vP2,vP3,g25)=veroneselImagesInG25(m4x2);
(degree pts,degree vP2,degree vP3,degree g25)
degree pts==n-1
"(L0,L1,L2,J)=adjunctionProcess(X,1)";
"L0_1==n and degree pts==n-1";
```

SeeAlso

```
adjunctionProcessData
aboRanestadSurface
///
```

doc ///

Key

```
aboRanestadSurface
(aboRanestadSurface, Ring, Number)
[aboRanestadSurface,Special]
[aboRanestadSurface,Smooth]
[aboRanestadSurface,Verbose]
```

Headline

find a random Abo-Ranestad surface

Usage

```
(X,m4x2) = aboRanestadSurface(P4,n)
```

Inputs

```
P4:Ring
the coordinate ring of P4
n:Number
the number of desired (-1) lines on the surface
```

Outputs

```
X:Ideal
the ideal of a Abo-Ranestad surface
m4x2:Matrix
a 4x2 matrix with linear entries over the exterior algebra
```

Description

Text

It the Tate resolution of an Abo-Ranestad surface there are a 4x2 matrix $m4x2$ and a 2x3 matrix $m2x3$ with linear entries over the exterior algebra. These matrices define rational maps $P3 \rightarrow G(2,5)$ and $P2 \rightarrow G(2,5)$ and the type of the surface depends on how these images intersect in the Grassmannian $G(2,5)$. It turns out that the number of

(-1) lines on the surface will coincide with the number of intersection points of the images + 1.

We need at least 3 intersection points and can have up to 7 intersection points.

In the construction we normalize the matrix $m2x3$ as indicated below

Example

```
kk=ZZ/nextPrime 10^3; e=symbol e; E=kk[e_0..e_4,SkewCommutative=>true];
m2x3=matrix{{e_0,e_1,e_3},{e_1,e_2,e_4}}
```

Text

One can easily force 3 or 4 intersection points. To find more we perform a random search over a finite ground field $\mathbb{F}_q$. Since an extra intersection point is a codimension 1 condition we can find examples with c additional intersection points with about q^c trials.

Example

```
P4=ZZ/nextPrime 10^3[x_0..x_4];
elapsedTime (X,m4x2)=aboRanestadSurface(P4,4);
minimalBetti X
singX=X+minors(2,jacobian X);
dim saturate singX==--1
(d,s)=(degree X, sectionalGenus X)
LeBarzN6(d,s,1)==8
Ksquare(d,s,1)==--12
elapsedTime betti (T=tateResolutionOfSurface X)
"elapsedTime (numList,L1,L2,J)=adjunctionProcess(X,3)";
"numList=={(4, 12, 13), 4, (12, 24, 13), 12, (12, 16, 5), 0, (4, 4, 1)}";
```

Text

The last adjoint surface is a Del Pezzo surface of degree 4 in $P4$. Thus

X is the blow-up in $12+9$ points embedded by a linear system of class $(12; 4^5, 2^{12}, 1^4)$.

Text

A special situation occurs when the 4×2 matrix $m_{4 \times 2}$ contains a 2×2 submatrix with entries in $e_0..e_2$ as well.

In that case we have two conics in the $e_0..e_2$ plane which intersect in 4 points, hence four intersection points in the Grassmannian $G(2,5)$.

We can force 2 more intersection points easily and can get a 7-th intersection point by a codimension 1 random search.

Example

```
elapsedTime (X,m4x2)=aboRanestadSurface(P4,7,Special=>2);
minimalBetti X
"elapsedTime (numList,L1,L2,J)=adjunctionProcess(X,3);";
"numList=={(4, 12, 13), 5, (12, 24, 13), 9, (12, 17, 6), 3, (5, 5, 1)}";
kk=ZZ/19;P4=kk[x_0..x_4];
setRandomSeed("fast search");
elapsedTime (X,m4x2)=aboRanestadSurface(P4,6,Special=>0,Verbose=>true);
minimalBetti X
"elapsedTime (numList,L1,L2,J)=adjunctionProcess(X,3);";
"numList=={(4, 12, 13), 6, (12, 24, 13), 6, (12, 18, 7), 6, (6, 6, 1)}";
setRandomSeed("another fast search");
elapsedTime (X,m4x2)=aboRanestadSurface(P4,8,Special=>2,Verbose=>true);
minimalBetti X
"elapsedTime (L0,L1,L2,J)=adjunctionProcess(X,3);";
"L0=={(4, 12, 13), 7, (12, 24, 13), 4, (12, 19, 8), 5, (7, 8, 2)}";
```

SeeAlso

adjunctionProcessData

///

doc ///

Key

matrixFromAboRanestadSurface

(matrixFromAboRanestadSurface, Ideal)

Headline

get the 4×2 matrix of an Abo-Ranestad surface

Usage

$m_{4 \times 2}$ = matrixFromAboRanestadSurface X

Inputs

X: Ideal

the ideal of a Abo-Ranestad surface

Outputs

$m_{4 \times 2}$: Matrix

a 4×2 matrix with linear entries over the exterior algebra

Description

Text

In the Tate resolution of an Abo-Ranestad surface there are a 4×2 matrix $m_{4 \times 2}$ and a 2×3 matrix $m_{2 \times 3}$ with linear entries over the exterior algebra. The 2×3 matrix is normalized. The function returns the 4×2 matrix.

Example

```
kk=ZZ/19
P4=kk[x_0..x_4]
setRandomSeed("fairly fast search")
elapsedTime (X,m4x2)=aboRanestadSurface(P4,6,Special=>2,Verbose=>true);
betty tateResolutionOfSurface X
elapsedTime X=aboRanestadSurfaceFromMatrix(P4,m4x2,Verbose=>true);
m4x2'=matrixFromAboRanestadSurface X
m4x2
assert(minors(2,sub(m4x2,vars P4))==minors(2,sub(m4x2',vars P4)))
```

SeeAlso

aboRanestadSurface

aboRanestadSurfaceFromMatrix

///

doc ///

Key

aboRanestadSurfaceFromMatrix

(aboRanestadSurfaceFromMatrix, Ring, Matrix)

[aboRanestadSurfaceFromMatrix, Smooth]

[aboRanestadSurfaceFromMatrix, Verbose]

Headline

get an Abo-Ranestad surface with given 4×2 matrix.

Usage

`X = aboRanestadSurfaceFromMatrix(P4,m4x2)`

Inputs

`P4:Ring`

the coordinate ring of `P4`

`m4x2:Matrix`

a 4x2 matrix with linear entries over the exterior algebra

Outputs

`X:Ideal`

the ideal of a Abo–Ranestad surface

Description

Text

In the Tate resolution of an Abo–Ranestad surface there are a 4x2 matrix `m4x2` and a 2x3 matrix `m2x3` with linear entries over the exterior algebra. These matrices define rational maps $P3 \rightarrow G(2,5)$ and $P2 \rightarrow G(2,5)$ and the type of the surface depends on how these images intersect in the Grassmannian $G(2,5)$. It turns out that the

number of

(–1) lines on the surface will coincide with the number of intersection points of the images + 1.

The function returns a corresponding surface `X`.

Example

`kk=ZZ/19`

`P4=kk[x_0..x_4]`

`setRandomSeed("fairly fast search")`

`elapsedTime (X,m4x2)=aboRanestadSurface(P4,6,Special=>2,Verbose=>true);`

`betti TateResolutionOfSurface X`

`elapsedTime X=aboRanestadSurfaceFromMatrix(P4,m4x2,Verbose=>true);`

`m4x2'=matrixFromAboRanestadSurface X`

`m4x2`

`assert(minors(2,sub(m4x2,vars P4))==minors(2,sub(m4x2',vars P4)))`

SeeAlso

`aboRanestadSurface`

`matrixFromAboRanestadSurface`

`///`

`doc ///`

Key

`specificAboRanestadSurface`

(`specificAboRanestadSurface`, `PolynomialRing`, `Ring`, `Number`)

Headline

Get the `k`-th specific Abo Ranestad surface

Usage

`(X,adjL)specificAboRanestadSurface(P4,E,k);`

Inputs

`P4:PolynomialRing`

coordinateRing of `P4`

`E:Ring`

the dual exterior algebra

`k:Number`

get example number `k`

Outputs

`X:Ideal`

the ideal of a Abo–Ranestad surface

`adjL:List`

precomputed adjunction data

Description

Text

In the Tate resolution of an Abo–Ranestad surface there is a 4x2 matrix `m4x2`.

We compute a Abo–Ranestad surface of `k`-th given matrix

Example

`kk=ZZ/19`

`P4=kk[x_0..x_4]`

`E=kk[e_0..e_4,SkewCommutative=>true]`

`elapsedTime (X,L0)=specificAboRanestadSurface(P4,E,2);`

`L0`

`"elapsedTime (numList,adjList,ptsList,J)=adjunctionProcess X;";`

`"numList==L0";`

`B=new BettiTally from {(0,{0},0) => 1, (1,{2},2) => 5, (2,{3},3) => 5, (3,{5},5) => 1}`

`"minimalBetti J == B";`

SeeAlso

`aboRanestadSurfaceFromMatrix`

`adjunctionProcess`

```

    tateResolutionOfSurface
    ///

doc ///
Key
get4x2Matrix
(get4x2Matrix, Matrix, Number)
[get4x2Matrix,Special]
Headline
get a 4x2 matrix for an Abo Ranestad surface construction
Usage
m4x2 = get4x2Matrix(m2x3,n)
Inputs
m2x3:Matrix
    the 2x3 matrix in the Tate resolution of the desired Abo–Ranestad surface
n:Number
    the desired number of intersection points in  $G(2,5)$ 
Outputs
m4x2:Matrix
    the 4x2 matrix of the Tate resolution of the desired Abo–Ranestad surface
Description
Text
    In the Tate resolution of an Abo–Ranestad surface there is a 4x2 matrix  $m4x2$  and a 2x3 matrix  $m2x3$ 
    with linear entries over the exterior algebra. These matrices define rational maps  $P3 \rightarrow G(2,5)$  and  $P2 \rightarrow G(2,5)$ 
    and the type of the surface depends on how these images intersect in the Grassmannian  $G(2,5)$ . It turns out that the
number of
    (-1) lines on the surface will coincide with the number of intersection points of the images + 1.
    The function returns for the normalized 2x3 matrix the desired 4x2 matrix.
Example
kk=ZZ/nextPrime 10^3
P4=kk[x_0..x_4]
E=kk[e_0..e_4,SkewCommutative=>true]
m2x3=matrix{{e_0,e_1,e_3},{e_1,e_2,e_4}}
m4x2=get4x2Matrix(m2x3,4)
elapsedTime X=aboRanestadSurfaceFromMatrix(P4,m4x2,Verbose=>true);
elapsedTime betti(T=tateResolutionOfSurface X)
"elapsedTime (numList,adjList,ptsList,J)=adjunctionProcess X;";
"numList=={(4, 12, 13), 5, (12, 24, 13), 9, (12, 17, 6), 3, (5, 5, 1)}";
B=new BettiTally from {(0,{0},0) => 1, (1,{2},2) => 5, (2,{3},3) => 5, (3,{5},5) => 1}
"minimalBetti J == B";
SeeAlso
aboRanestadSurfaceFromMatrix
adjunctionProcess
tateResolutionOfSurface
    ///

    ///
ci=ideal(gens X*random(source gens X,P4^{2:-5}));
Y=ci:X;
minimalBetti Y
betti tateResolutionOfSurface Y
(d,sg)=(degree Y, sectionalGenus Y)
Ksquare(d,sg,4)
D1=canonicalDivisor Y;
D2=canonicalDivisor Y;
D3=canonicalDivisor Y;
baseLocus=saturate(D1+D2+D3)
degree baseLocus, genus baseLocus
R=residualInQuintics X;
R==baseLocus
degree baseLocus, genus baseLocus, selfIntersectionNumber(Y,baseLocus)
E1=D1:baseLocus;
dim E1, degree E1, genus E1, selfIntersectionNumber(Y,E1)
    ///

--* Abo surfaces *-
doc ///
Key

```

aboSurfaceFromMatrix

(aboSurfaceFromMatrix,Matrix,Ring)

[aboSurfaceFromMatrix,Verbose]

Headline

construct a Abo surface, a K3 surface of degree 12 and sectional genus 13

Usage

X= aboSurfaceFromMatrix(m3x4,P4)

Inputs

m3x4:Matrix

a 3x4 matrix with linear entries in the exterior algebra E

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of a Abo surface

Description

Text

In the Tate resolution of Abo surfaces there are linear 3x1 and linear 3x4 matrices.

We assume that $\text{transpose } m3x1 = \text{matrix}\{e_0, e_1, e_2\}$. Whether the given m3x4 matrix together with m3x1 leads to a smooth surface can be tested with testMatrix1.

The resulting surfaces are all K3-surfaces blown-up in 6 points and the degree of the six exceptional divisor form a partition of 12 into 6 parts. In the example below this is the partition {1,2,2,2,3}.

Example

kk=ZZ/19

P4=kk[x_0..x_4];

E=kk[e_0..e_4,SkewCommutative=>true]

m3x4=matrix{{7*e_0+3*e_1-7*e_2-8*e_3, -4*e_0+3*e_1-7*e_2-8*e_4, 5*e_0+6*e_1+2*e_2+9*e_3-4*e_4, -6*e_0-7*e_1+3*e_3-3*e_4}, {e_0-5*e_1-e_2+7*e_3, -e_0-4*e_1+7*e_2-2*e_4, -e_0-8*e_1+2*e_3-3*e_4, -7*e_0+3*e_1-9*e_2+6*e_3-6*e_4}, {8*e_0-2*e_1+3*e_2+6*e_3, 9*e_0-2*e_1-e_2-e_4, 9*e_0-9*e_1+7*e_2+e_3+8*e_4, -e_0+7*e_1-8*e_2+8*e_3-8*e_4}}

elapsedTime X=aboSurfaceFromMatrix(m3x4,P4);

(d,sg)=(degree X, sectionalGenus X)

B=betti tateResolutionOfSurface X

xO=chiO(X)

Ksquare(d,sg,xO)=-6

HdotK(d,sg)=12

LeBarzN6(d,sg,xO)

minimalBetti X

residual=residualInQuintics X;

dim residual, degree residual

cResidual=primaryDecomposition residual;

tally apply(cResidual,c->(dim c, degree c, dim (c+X), degree (c+X)))

D=canonicalDivisor X;

degree D

selfIntersectionNumber(X,D)

cD=primaryDecomposition D;#cD

tally apply(cD,c->(Oc=sheaf(P4^1/c); r=rank HH^0(Oc);

(dim c,degree c,r)))

partits=partitionOfCanonicalDivisorOfAboSurface(X,Verbose=>true)

Text

X=X_min(p1..p6) is a minimal K3 surface blown up in 6 points embedded by a linear system $H = |(Hmin;3,2^4,1)|$. The four (-1) conic decompose into two Frobenius orbits of length 2 and 2.

SeeAlso

partitionOfCanonicalDivisorOfAboSurface

canonicalDivisor

selfIntersectionNumber

chiO

Ksquare

residualInQuintics

LeBarzN6

///

doc ///

Key

testMatrix

(testMatrix,Matrix,Ring)

[testMatrix,SingX]

```
[testMatrix,Verbose]
testMatrix1
(testMatrix1,Matrix,Ring)
[testMatrix1,SingX]
[testMatrix1,Verbose]
testMatrix2
(testMatrix2,Matrix,Ring)
[testMatrix2,Verbose]
[testMatrix2,WithM3x13]
[testMatrix2,WithX]
```

Headline

test whether a 3x4 matrix with entries over the exterior algebra leads to an Abo surface

Usage

```
r = testMatrix1(m3x4,P4)
singX = testMatrix(m3x4,P4)
```

Inputs

m3x4:Matrix
a 3x4 matrix with linear entries in the exterior algebra E
P4:PolynomialRing
coordinate ring of P4

Outputs

r:ZZ
the rank of the crucial Hom space
singX: Ideal
ideal of the singular locus of an example of a surfaces from m3x4

Description

Text

In the Tate resolution of Abo surfaces there are linear 3x1 and linear 3x4 matrices.
We assume that transpose m3x1= matrix{{e_0,e_1,e_2}}. Whether the given m3x4 matrix together with m3x1 leads to a smooth surface can be tested with testMatrix1.
We need that r1>5.

Example

```
kk=ZZ/19
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true]
m3x1=transpose matrix{{E_0,E_1,E_2}}
m3x4=matrix {{7*e_0+3*e_1-7*e_2-8*e_3, -4*e_0+3*e_1-7*e_2-8*e_4, 5*e_0+6*e_1+2*e_2+9*e_3-4*e_4,
-6*e_0-7*e_1+3*e_3-3*e_4}, {e_0-5*e_1-e_2+7*e_3, -e_0-4*e_1+7*e_2-2*e_4, -e_0-8*e_1+2*e_3-3*e_4,
-7*e_0+3*e_1-9*e_2+6*e_3-6*e_4}, {8*e_0-2*e_1+3*e_2+6*e_3, 9*e_0-2*e_1-e_2-e_4,
9*e_0-9*e_1+7*e_2+e_3+8*e_4, -e_0+7*e_1-8*e_2+8*e_3-8*e_4}}
elapsedTime r1=testMatrix1(m3x4,P4)
elapsedTime r2=testMatrix2(m3x4,P4)
r1==r2+5
elapsedTime singX=testMatrix(m3x4,P4)
```

Text

The matrix m3x4 give rise to a surface if r>5.

Example

```
X= aboSurfaceFromMatrix(m3x4,P4);
beti tateResolutionOfSurface X
elapsedTime singX=testMatrix(m3x4,P4)
```

Text

The last function also checks whether there is a smooth surface with these matrices.
For a general 3x4 matrix we have r=5.

Example

```
setRandomSeed("really general");
m3x4g=random(E^3,E^{4:-1});
testMatrix1(m3x4g,P4)==5
```

Text

The second version checks whether the matrix can lead to a smooth surface
testMatrix(m3x4g,P4) -- does not seem to return anything

SeeAlso

aboSurfaceFromMatrix

```
///
///
B=betti(hom=Hom(coker m3x4,coker m3x1**E^{1},DegreeLimit=>2))
r=if member((0,{0},0),keys B) then B#(0,{0},0) else 0
kk=coefficientRing ring m3x4
c:=null;
genHomo=sum(r,i->(
while (c=random kk; c==0) do ();c*hom_i));
```

```

m3x3=homomorphism genHomo
betti (m3x4=matrix(m3x1**E^{1}|m3x3))
betti (s1=syz m3x4)
s2=s1*((id_(E^{4:-1}))|map(E^{5:-2},E^{4:-2},0))|random(source s1,E^{4:-2}))
betti (T1=res coker transpose s2)
betti (T2=res(coker transpose T1.dd_6,LengthLimit=>8))
X=saturate ideal syz symExt(T2.dd_8,P4);
minimalBetti X
dim X==3 and degree X==12 and sectionalGenus X ==13
needsPackage "FastMinors"
viewHelp "FastMinors"
elapsedTime regularInCodimension(2,P4/X)
elapsedTime (singX=singularLocus X; dim singX==0)

///

doc ///
Key
analyzeAboSurface
(analyzeAboSurface,Ideal)
(analyzeAboSurface,Ring)
(analyzeAboSurface,ZZ,Ring)
[analyzeAboSurface,PrintConstructionData]
[analyzeAboSurface,Verbose]
Headline
analyze Abo surface
Usage
(K,residual) = analyzeAboSurface X
(K,residual) = analyzeAboSurface P4
(K,residual) = analyzeAboSurface(n,P4)
Inputs
X:Ideal
  of an Abo surface
P4:PolynomialRing
  coordinate ring of P4
n: ZZ
  number of random Abo surfaces to be analyzed
Outputs
K:List
  the partition of the canonical divisor of X
residual: Ideal
  the residualInQuintics ideal
Description
Text
  We analyze the residualInQuintics scheme and the partition of the canonical divisor of X
  in view of Le Barz 6-secant formula.
Example
kk=ZZ/nextPrime 10^4;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
setRandomSeed("fix decompositions");
elapsedTime (X,m3x4)=abo111333Surface(P4,E,Verbose=>false);
elapsedTime (K,residual)=analyzeAboSurface(X,Verbose=>false);
K
cResidual=primaryDecomposition residual;
tally apply(cResidual, c-> (dim c, degree c, betti c, dim(c+X), degree (c+X),
  tally apply(primaryDecomposition(c+X),d->(dim d, degree d, degree radical d))))
(d,sg,xO)=(12,13,2);
Ksquare(d,sg,xO) == -#K
numberOfMinusOneLines=#select(K,d->d==1)
numberOfSixSecants=sum(select(cResidual,c->dim c == 2 and degree (c+X)==6),d->degree d)
LeBarzN6(d,sg,xO)==numberOfMinusOneLines+numberOfSixSecants
Text
  In this specific example X has four 6-secant lines. The intersection of these lines
  with X decomposes into Frobenius orbits of length (1,5) (twice), length (1,1,2,2)
  and length (6) respectively.
Example
R=(select(cResidual,c->degree c==4))_0;-- a rational normal curve of degree 4
minimalBetti R

```

```

    saturate ideal singularLocus R
    degree (R+X)==21
Text
    Since the rational normal curve R intersects X in  $21 > 20$  points,
    it is contained in any quintic of X.
SeeAlso
    LeBarzN6
    residualInQuintics
///

doc ///
Key
    partitionOfCanonicalDivisorOfAboSurface
    (partitionOfCanonicalDivisorOfAboSurface,ideal)
    [partitionOfCanonicalDivisorOfAboSurface,Equations]
    [partitionOfCanonicalDivisorOfAboSurface,Verbose]
Headline
    compute the partition of the canonical divisor
Usage
    K = partitionOfCanonicalDivisorOfAboSurface X
    IK = partitionOfCanonicalDivisorOfAboSurface(X,Equations=>true)
Inputs
    X: Ideal
        of an Abo surface
Outputs
    K: List
        the partition of the canonical divisor of X
    IK: Ideal
        the homogeneous ideal of the canonical divisor
Description
Text
    The canonical divisor of a Abo surface is a collection of six  $(-1)$  curves
    of total degree 12. Which degrees occur depends on the surface.
Example
    kk=ZZ/nextPrime 10^4;
    P4=kk[x_0..x_4];
    E=kk[e_0..e_4,SkewCommutative=>true];
    setRandomSeed("fix decompositions");
    elapsedTime (X,m3x4)=abo111333Surface(P4,E,Verbose=>false);
    elapsedTime K = partitionOfCanonicalDivisorOfAboSurface X
    (d,sg,xO)=(12,13,2);
    Ksquare(d,sg,xO) , -#K
    HdotK(d,sg), sum K
Text
    The partitions of 12 into 6 parts are in bijections with partitions of 6. From the
    11 partitions we have observed the following six
Example
    #partitions(6)
    #{{1, 1, 1, 1, 1, 7}, {1, 1, 1, 2, 3, 4}, {1, 1, 1, 3, 3, 3}, {1, 1, 2, 2, 2, 4},
    {1, 1, 2, 2, 3, 3}, {1, 2, 2, 2, 2, 3}}
Text
    In some cases the 6 points blown-up are infinitesimal near. We
    think that these are boundary cases, which hence give no new families.
SeeAlso
    abo111333Surface
    abo111117Surface
    Ksquare
    HdotK
///

doc ///
Key
    abo111333Surface
    (abo111333Surface, Ring, Ring)
    [abo111333Surface, Verbose]
Headline
    get an Abo surface whose canonical divisors partitions into components of degrees {1,1,1,3,3,3}
Usage

```

(X,m3x4) = abo111333Surface(P4,E)

Inputs

P4: [Ring](#)
coordinate [ring](#) of P4
E: [Ring](#)
[dual](#) exterior algebra

Outputs

X: [Ideal](#)
[ideal](#) of an Abo surface X
m3x4: [Matrix](#)
the 3x4 [matrix](#) of linear forms over the exterior algebra

Description

Text

This gives an (apparently) unirational construction of Abo surfaces with 111333 [partition](#) of the canonical divisor. Add how it works.

Example

```
kk=ZZ/nextPrime 10^4;  
P4=kk[x_0..x_4];  
E=kk[e_0..e_4,SkewCommutative=>true];  
setRandomSeed("fix decompositions");  
elapsedTime (X,m3x4)=abo111333Surface(P4,E,Verbose=>false);  
elapsedTime (K,residual)=analyzeAboSurface(X,Verbose=>false);  
K  
cResidual=primaryDecomposition residual;  
tally apply(cResidual, c-> (dim c, degree c, betti c, dim(c+X), degree (c+X),  
    tally apply(primaryDecomposition(c+X),d->(dim d, degree d, degree radical d))))  
(d,sg,xO)=(12,13,2);  
Ksquare(d,sg,xO) == -#K  
numberOfMinusOneLines=#select(K,d->d==1)  
numberOfSixSecants=sum(select(cResidual,c->dim c == 2 and degree (c+X)==6),d->degree d)  
LeBarzN6(d,sg,xO)==numberOfMinusOneLines+numberOfSixSecants
```

Text

In this specific example X has four 6-secant [lines](#). The [intersection](#) of these [lines](#) with X decomposes into Frobenius orbits of [length](#) (1,5) (twice), [length](#) (1,1,2,2) and [length](#) (6) respectively.

Example

```
R=(select(cResidual,c->degree c==4))_0;-- a rational normal curve of degree 4  
minimalBetti R  
saturate ideal singularLocus R  
degree (R+X)==21
```

Text

Since the rational normal curve R intersects X in $21 > 20$ points, it is contained in any quintic of X.

SeeAlso

LeBarzN6
residualInQuintics
partitionOfCanonicalDivisorOfAboSurface
analyzeAboSurface
aboSurfaceFromMatrix

///

doc ///

Key

abo111117Surface
(abo111117Surface,[Ring](#),[Ring](#))
[abo111117Surface,[Verbose](#)]

Headline

[get](#) an Abo surface whose canonical divisors [partitions](#) into [components of degrees](#) {1,1,1,1,1,7}

Usage

(X,m3x4) = abo111117Surface(P4,E)

Inputs

P4: [Ring](#)
coordinate [ring](#) of P4
E: [Ring](#)
[dual](#) exterior algebra

Outputs

X: [Ideal](#)
[ideal](#) of an Abo surface X

m3x4: **Matrix**
the 3x4 **matrix of** linear forms over the exterior algebra

Description

Text

This gives an (apparently) unirational construction of Abo surfaces with 111117 **partition** of the canonical divisor. Add how it works.

Example

```
kk=ZZ/nextPrime 10^4;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
setRandomSeed("fix a fast decomposition of K");
elapsedTime (X,m3x4)=abo111117Surface(P4,E,Verbose=>false);
elapsedTime (K,residual)=analyzeAboSurface(X,Verbose=>false);
K
cResidual=primaryDecomposition residual;
netList apply(cResidual, c->(dim c, degree c, betti c, dim(c+X), degree (c+X),
    tally apply(primaryDecomposition(c+X),d->(dim d, degree d, degree radical d))))
(d,sg,xO)=(12,13,2);
Ksquare(d,sg,xO) == -#K
numberOfMinusOneLines=#select(K,d->d==1)
expectedNumberOfSixSecants=LeBarzN6(d,sg,xO)-numberOfMinusOneLines
plane=cResidual_0
cPlaneCapX=primaryDecomposition saturate(plane+X)
point=(select(cPlaneCapX,c->dim c==1))_0
randomLineThroughPoint=trim(plane+ideal ((gens point)*random(source gens point,P4^(-1))))
degree(randomLineThroughPoint+X)==6
L1=cResidual_4
degree(L1+X)
dim(L1+plane)
tally apply(cResidual_{1,2,3},L->dim(L+plane))
```

Text

In this example X has a pencil of 6-secant **lines**:

All the **lines** in the plane trough the point. Thus LeBarz 6-secant formula does **not apply**.

There are three further 6-secants **lines** one of which is L1.

The 5-secant **lines** are contained in every quintic, because each intersects the plane in a point.

SeeAlso

LeBarzN6
partitionOfCanonicalDivisorOfAboSurface
analyzeAboSurface
///

doc ///

Key

specificAboSurface
(specificAboSurface, **Ring**, **Ring**, **Number**)
[specificAboSurface, **Verbose**]

Headline

compute a smooth Abo surface of given data (m3x4,d,(K,R))

Usage

X = specificAboSurface(P4,E,k)
X = specificEllipticAboSurface(P4,E,k)

Inputs

P4:**Ring**
coordinate **ring of** P4 over a ground field of characteristic 3

E: **Ring**
exterior algebra **dual to** P4.

k: **Number**
a **number** which specifies the m3x4 **matrix of** linear forms over E to use.

Outputs

X:**Ideal**
ideal of a Abo surface

Description

Text

In characteristic p=19 the function returns one of 7 specific Abo surface corresponding to one of the following **partitions** of the canonical divisor:

{1, 2, 2, 2, 2, 3}, {1, 1, 2, 2, 3, 3}, {1, 1, 2, 2, 2, 4}, {1, 1, 1, 3, 3, 3},

{1, 1, 1, 2, 3, 4}, {1, 1, 1, 2, 2, 5}, {1, 1, 1, 1, 1, 7}.

Other cases are $p=11$ and $p=7$. In characteristic 7 the example with $k=1$ has the partition

$\{1, 1, 1, 1, 2, 6\}$.

Example

```
kk=ZZ/19;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
X=specificAboSurface(P4,E,1);
minimalBetti X
(d,sg)=(degree X, sectionalGenus X)
Ksquare(d,sg,2)=-6
LeBarzN6(d,sg,2)=7
pK=partitionOfCanonicalDivisorOfAboSurface X
RX=residualInQuintics X;dim RX
RX==radical RX
tally apply(decompose RX,c->(dim c ,degree c, genus c,(dim (c+X),degree(c+X))))
```

Text

There are two (-1) lines and five 6-secant lines in this case.

The Tate resolution has a specific form

Example

```
beti(T=tateResolutionOfSurface X)
m3x4=T.dd_5^{0..2}_{4..7}
m3x1=(transpose gens trim ideal T.dd_4^{0..2}_{3})**(ring T)^{-2}
beti(hom=Hom(coker m3x4,coker m3x1,DegreeLimit=>0))
```

Text

The construction of X uses a special 3×4 matrix over E such that the Hom group above is non zero.

The $(5,5)$ linked surface Y is an elliptic surface of degree 13 and sectional genus 16.

Example

```
setRandomSeed("get a Smooth Surface");
beti(ci=ideal (gens X*random(source gens X,P4^{2:-5})))
minimalBetti (Y=ci:X)
RY=residualInQuintics Y;
degree RY
RY==canonicalDivisor X
saturate ideal singularLocus (P4/Y)== ideal 1_P4
D1=canonicalDivisor Y;
D2=canonicalDivisor Y;
beti(baseLocus=saturate (D1+D2))
degree baseLocus, genus baseLocus
selfIntersectionNumber(Y,baseLocus)
(degree Y, sectionalGenus Y)=(13,16)
beti tateResolutionOfSurface Y
Ksquare(13,16,3)=-5
LeBarzN6(13,16,3)=LeBarzN6(12,13,2)
beti(E1=D1:baseLocus)
degree E1, genus E1, selfIntersectionNumber(Y,E1)
```

Text

The linked surface Y is a smooth elliptic surface blowup in 5 points which are (-1) lines on Y .

Example

```
kk=ZZ/31;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
X=specificAboSurface(P4,E,1,Verbose=>true);
minimalBetti X
```

Text

This surface is a non-minimal Elliptic surface with four (-1) lines and two (-1) conics. The canonical divisor has in addition a degree 4 elliptic curve as a component.

SeeAlso

```
residualInQuintics
tateResolutionOfSurface
selfIntersectionNumber
LeBarzN6
Ksquare
canonicalDivisor
tateResolutionOfSurface
```

///

```

doc ///
Key
specificEllipticAboSurfaceD12S13
(specificEllipticAboSurfaceD12S13, Ring, Ring, Number)
[specificEllipticAboSurfaceD12S13, Verbose]
Headline
compute a specific smooth elliptic Abo surface of given data (m3x4,d,(K,R))
Usage

X = specificEllipticAboSurfaceD12S13(P4,E,k)
Inputs
P4:Ring
coordinate ring of P4 over a ground field of characteristic 3
E: Ring
exterior algebra dual to P4.
k: Number
a number which specifies the m3x4 matrix of linear forms over E to use.
Outputs
X:Ideal
ideal of a Abo surface
Description
Text
In characteristic p=31 the function returns an non minimal elliptic surface with
six (-1) curves of degree {1,1,1,1,2,2}.
The canonical divisor has in addition a component which is an elliptic curve of degree 4.
Example
kk=ZZ/31;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
X=specificEllipticAboSurfaceD12S13(P4,E,0,Verbose=>true);
minimalBetti X
Text
This surface is an non-minimal Elliptic surface with four (-1) lines and two (-1) conics.
The canonical divisor has in addition a degree 4 elliptic curve as a component.
SeeAlso
residualInQuintics
tateResolutionOfSurface
selfIntersectionNumber
LeBarzN6
Ksquare
canonicalDivisor
tateResolutionOfSurface
///

///

kk=ZZ/19;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
X=specificAboSurface(P4,E,4);
minimalBetti X
saturate ideal singularLocus(P4/X)
(d,sg)=(degree X, sectionalGenus X)
Ksquare(d,sg,2)=-6
LeBarzN6(d,sg,2)=7
beti(T=tateResolutionOfSurface X)
beti(ci=ideal (gens X*random(source gens X,P4^{2:-5})))
minimalBetti (Y=ci:X)
RX=residualInQuintics X;
RY=residualInQuintics Y;
planes=select(decompose RX,c-->dim c==3)
minimalBetti(Y1=Y:planes_0)
singY1=saturate ideal singularLocus(P4/Y1)
dim singY1, degree singY1
decompose singY1
decompose Y1

```

```

degree RY
RY==canonicalDivisor X
saturate ideal singularLocus (P4/Y)== ideal 1_P4
D1=canonicalDivisor Y;
D2=canonicalDivisor Y;
betti(baseLocus=saturate (D1+D2))
degree baseLocus, genus baseLocus
selfIntersectionNumber(Y,baseLocus)
(degree Y, sectionalGenus Y)==(13,16)
betti tateResolutionOfSurface Y
Ksquare(13,16,3)==-5
LeBarzN6(13,16,3)==LeBarzN6(12,13,2)
betti(E1=D1:baseLocus)
degree E1, genus E1, selfIntersectionNumber(Y,E1)

tally apply(decompose RY,c->(dim c-1, degree c,genus c,
(dim(c+Y)-1, degree(c+Y)),selfIntersectionNumber(X,c)))
tally apply(select(decompose baseLocus,c->dim c ==2),c->(dim c -1, degree c, genus c,
selfIntersectionNumber(Y,c)))
betti(E2=D2:baseLocus)
degree E2, genus E2, selfIntersectionNumber(Y,E2)
residualInQuintics X==baseLocus
dim (E1+E2), degree(E1+E2)
///

doc ///
Key
collectAboSurfaces
(collectAboSurfaces, List, Ring, Ring, ZZ)
[collectAboSurfaces,Verbose]
[collectAboSurfaces,Special]
Headline
collect Abo surfaces
Usage
mdKR'= specificAboSurfaces(mdKRs,P4,E,N)
Inputs
mdKRs: List
  of Abo surface data m3x4,d,(K,R)
P4:Ring
  coordinate ring of P4 over a small finite ground field
E: Ring
  exterior algebra dual to P4.
N: ZZ
  the desired total number of surface data.
Outputs
mdKR':List
  of Abo surface data m3x4,d,(K,R)
Description
Text
  The functions collects N surface data by choosing randomly 3x4 matrices over the exterior
  algebra and testing whether they lead to a surface.
  If the pair of (K,R) of partition of the canonical divisor and numerical type of
  the residualInQuintics scheme is new or of that type there are only few examples in the list then the
  the example will be appended to the current list. The functions stops if the total number N of
  examples is reached.

  In case of the option Special=>true, the m3x4 Bordiga matrix has a rank 1 point.
Example
kk=ZZ/19;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
mdKRs={};
setRandomSeed("carefully choosen randomSeed");
elapsedTime mdKRs'=collectAboSurfaces(mdKRs,P4,E,1) -- 31.8852s elapsed

SeeAlso
residualInQuintics
partitionOfCanonicalDivisorOfAboSurface

```

```

///

/// --prepare specificAboExample
kk=ZZ/11
P4=kk[x_0..x_4]
E=kk[e_0..e_4,SkewCommutative=>true]
elapsedTime minimalBetti (X=specificAboSurface(P4,E,9))

kk=ZZ/5
P4=kk[x_0..x_4]
E=kk[e_0..e_4,SkewCommutative=>true]
setRandomSeed("start from here");
mdKRs={}
elapsedTime mdKRs'=collectAboSurfaces(mdKRs,P4,E,20,Special=>true);#mdKRs'
elapsedTime mdKRs=collectAboSurfaces(mdKRs',P4,E,50);#mdKRs
#mdKRs
mdKRs= reverse sort(mdKRs , mdKR-> mdKR_2_0);
Ta=tally apply(mdKRs , mdKR-> mdKR_2_0)
netList apply(mdKRs,mdKR->(mdKR_1,mdKR_2))
pos=apply(reverse sort keys Ta,K->position(mdKRs,mdKR->K==mdKR_2_0))
toString mdKRs_pos
-- 1) use cut and paste into specificAboSurface
-- 2) replace e->E use replace nEw -> new
-- 3) adapt the coments and error handling in specificAboSurface
#pos
kk=ZZ/7;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
elapsedTime minimalBetti (X=specificAboSurface(P4,E,4))
elapsedTime apply(7,k->minimalBetti (X=specificAboSurface(P4,E,k)))

```

```

///

```

```

doc ///
Key
randomAboSurface
randomSpecialAboSurface
randomAboSurfaceWithLargeHomSpace
randomAboSurfaceWithHomSpaceOfGivenDimension
(randomAboSurface, Ring, Ring)
[randomAboSurfaceWithLargeHomSpace, Verbose]
[randomAboSurfaceWithLargeHomSpace, Count]
[randomAboSurfaceWithHomSpaceOfGivenDimension, Verbose]
[randomAboSurfaceWithHomSpaceOfGivenDimension, Count]
[randomAboSurface, Verbose]
[randomAboSurface, Count]
(randomSpecialAboSurface, Ring, Ring)
[randomSpecialAboSurface, Verbose]
[randomSpecialAboSurface, Count]
[randomSpecialAboSurface, PrintConstructionData]
[randomAboSurface, PrintConstructionData]
Headline
get random Abo surfaces
Usage
(X,m3x4)=randomAboSurface(P4,E)
(X,m3x4)=randomSpecialAboSurface(P4,E)
(X,m3x4,r)=randomAboSurfaceWithLargeHomSpace(P4,E,h)
(X,m3x4,r)=randomAboSurfaceWithLargeHomSpace(P4,E,h)
Inputs
P4:Ring
coordinate ring of P4 over a small finite ground field.
E: Ring
exterior algebra dual to P4
h: ZZ
dimension of the desired Hom-space, or a lower bound for that
Outputs
X:Ideal
of an Abo surface
m3x4: Matrix

```

3x4 **matrix** over the exterior algebra
r:ZZ
dimension of the **Hom**-space

Description

Text

The functions constructs a Abo surface by choosing randomly 3x4 matrices over the exterior algebra and testing whether they lead to a surface.

In the case of randomSpecialAboSurface, the m3x4 Bordiga **matrix** has a **rank** 1 point.

The two more special **version** search for Abo surface with lower bound or precise dimension of the **Hom** space. These are search functions which **take** long.

Example

```
kk=ZZ/7;  
P4=kk[x_0..x_4];  
E=kk[e_0..e_4,SkewCommutative=>true];  
setRandomSeed("carefully choosen fast randomSeed");  
elapsedTime (X,m3x4)=randomAboSurface(P4,E);  
saturate minors(2,sub(m3x4,vars P4))  
setRandomSeed("same start");  
elapsedTime (Xs,m3x4s)=randomSpecialAboSurface(P4,E);  
pt=saturate minors(2,sub(m3x4s,vars P4))  
sub(m3x4s,vars P4)%pt
```

SeeAlso

residualInQuintics
partitionOfCanonicalDivisorOfAboSurface

///

doc ///

Key

randomEllipticAboSurface
(randomEllipticAboSurface, **Ring**, **Ring**)
[randomEllipticAboSurface, **Verbose**]
[randomEllipticAboSurface, **Count**]
[randomEllipticAboSurface, **PrintConstructionData**]
[randomEllipticAboSurface, **NumberOfRank1Points**]

Headline

Get **random** elliptic Abo surface

Usage

(X,m3x4,r)=randomEllipticAboSurface(P4,P3)

Inputs

P4: **Ring**
coordinate **ring** of P4 over a small finite ground field
P3: **Ring**
coordinate **ring** of a P3

Outputs

X: **Ideal**
of an Abo surface
m3x4: **Matrix**
over the exterior algebra **dual** P4
r: **ZZ**
rank of solution space, dimension of the **Hom** space is 115-r

Description

Text

The functions constructs a **random** elliptic Abo surface by choosing carefully 3x5 over P3 such that the **adjoint** Bordiga **matrix** has the desired **number of rank** 1 points.

The fastest hence **default** choice is NumberOfRank1Points=>3.

Example

```
kk=ZZ/19;  
P4=kk[x_0..x_4];  
P3=kk[y_0..y_3];  
setRandomSeed("fairly fast");  
elapsedTime (X,m3x4,r)=randomEllipticAboSurface(P4,P3);  
minimalBetti X  
(d,sg)=(degree X, sectionalGenus X)  
Ksquare(d,sg,2)
```

```

beti(tateResolutionOfSurface X)
RX=residualInQuintics X;
cRX=decompose RX;
tally apply(cRX, c-> (dim c -1, degree c, (dim(c+X)-1,degree(c+X))))
K=canonicalDivisor X;
cK=decompose K;
tally apply(cK,c->(dim c -1, degree c, genus c, selfIntersectionNumber(X,c), minimalBetti c))

```

Text

The canonical divisor decomposes into four (-1) lines and two $(1-)$ conics, and elliptic curve of class $(2,2)$ on a $P^1 \times P^1$. So the minimal model has $K^2_{\min}=0$, hence is an elliptic surface.

Example

```

E=(select(cK,c->genus c ==1))_0;
beti E,
saturate ideal singularLocus(P4/E)
dim E,degree E, genus E, minimalBetti E
m=2
mE= saturate(E^m+X);
beti mE
H=ideal( gens mE*random(source gens mE,P4^{6}));
linked=saturate((H+X):mE);
beti linked, beti saturate linked
H'=ideal(gens linked*random(source gens linked,P4^{6}));
beti trim ideal(gens linked%X)
beti linked, beti X
E'=(X+H'):linked;
beti E'
degree E', genus E'
dim(E'+E)

```

Text

the divisor $2E$ moves in a pencil.

SeeAlso

residualInQuintics

///

—* classical rational surfaces *—

doc ///

Key

cubicScroll
(cubicScroll, PolynomialRing)

Headline

construct the cubic scroll

Usage

X= cubicScroll P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal
of the cubic scroll in P4

Description

Text

The smooth cubic scroll is uniquely determined up to projectivities.

It is define by the 2×2 minors a 2×3 matrix.

The function returns this ideal

Example

```

kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=cubicScroll P4
minimalBetti X

```

///

doc ///

Key

bordigaSurface
(bordigaSurface, PolynomialRing)

Headline

construct a Bordiga surface

Usage

X= bordigaSurface P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of the Bordiga surface in P4

Description

Text

Constructs a Bordiga surface via its Hilbert–Burch matrix

Example

```
kk=ZZ/nextPrime 10^3;
```

```
P4=kk[x_0..x_4];
```

```
minimalBetti(X=bordigaSurface P4)
```

```
///
```

```
doc ///
```

Key

veroneseSurface

(veroneseSurface, PolynomialRing, PolynomialRing)

Headline

construct the Veronese surface

Usage

X= veroneseSurface(P4,P2)

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of the Veronese surface in P4

Description

Text

We compute the image from the parameterization.

Example

```
kk=ZZ/nextPrime 10^3;
```

```
P4=kk[x_0..x_4];
```

```
P2=kk[y_0..y_2]
```

```
minimalBetti(X=veroneseSurface(P4,P2))
```

```
///
```

```
doc ///
```

Key

delPezzoSurface

(delPezzoSurface, PolynomialRing)

Headline

construct the del Pezzo surface

Usage

X= delPezzoSurface P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of the cubic scroll in P4

Description

Text

We choose randomly two quadrics.

Example

```
kk=ZZ/nextPrime 10^3;
```

```
P4=kk[x_0..x_4];
```

```
minimalBetti(X=delPezzoSurface P4)
```

```
///
```

```
doc ///
```

Key

castelnuovoSurface

(castelnuovoSurface, PolynomialRing)

Headline

construct the castelnuovo surface

Usage

X= castelnuovoSurface P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal

of the del Pezzo in P4

Description

Text

We construct a Castelnuovo surface from its Hilbert–Burch matrix

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];  
minimalBetti(X=castelnuovoSurface P4)  
(d,sg)=(degree X, sectionalGenus X)  
Ksquare(5,2,1)
```

Text

X is a conic bundle. We compute the number of singular fibers.

Example

```
P1=kk[z_0,z_1]  
P4xP1=P4**P1  
fibration=ideal(sub(syz gens X,P4xP1) * transpose sub(vars P1,P4xP1));  
singFibration= fibration+minors(3,diff(transpose sub(vars P4,P4xP1), gens fibration));  
singularFibers=trim sub(saturate(singFibration,ideal sub(vars P4,P4xP1)),P1)  
degree singularFibers
```

Text

The are 7 singular fibers,
which fits with Ksquare=1==8-7

///

doc ///

Key

okonekSurfaceD8S6

(okonekSurfaceD8S6, PolynomialRing, Ring)

Headline

construct a degree 8 Okonek surface, sectional genus 6

Usage

X=okonekSurfaceD8S6(P4,E)

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal

of a degree 8 surface

Description

Text

We construct the surface from a randomly choosen differential T.dd_3
of the Tate resolution of the desired ideal.

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];  
E=kk[e_0..e_4,SkewCommutative=>true];  
minimalBetti(X=okonekSurfaceD8S6(P4,E))  
degree X, sectionalGenus X  
beti(T=tateResolutionOfSurface X)  
beti T.dd_3  
ci=ideal(gens X*random(source gens X,P4^{-3,-4}));  
Y=ci:X;  
minimalBetti Y  
P2=kk[y_0..y_2];  
minimalBetti veroneseSurface(P4,P2)
```

///

doc ///

Key

ionescuOkonekSurfaceD7

(ionescuOkonekSurfaceD7, PolynomialRing,PolynomialRing)

Headline

construct a Ionescu–Okonek surface of degree 7 and sectional genus 4

Usage

```
X= ionescuOkonekSurfaceD7(P4,P2)
```

Inputs

```
P4:PolynomialRing
  coordinate ring of P4
P2:PolynomialRing
  coordinate ring of P2
```

Outputs

```
X:Ideal
  of a Ionescu–Okonek surface in P4
```

Description

Text

We construct a Ionescu–Okonek surface from its rational parametrization.

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
P2=kk[y_0..y_2];
minimalBetti(X=ionescuOkonekSurfaceD7(P4,P2))
degree X, sectionalGenus X

///
///
apply((toList(-4..8),m->chil(m,8,5,1)))
m2x2=random(E^2:2},E^2:1})
betty (T=res coker m2x2)
///
```

doc ///

Key

```
ionescuOkonekSurfaceD8S5
(ionescuOkonekSurfaceD8S5, PolynomialRing,PolynomialRing)
```

Headline

construct a Ionescu–Okonek surface of degree 8 and sectional genus 5

Usage

```
X= ionescuOkonekSurfaceD8S5(P4,P2)
```

Inputs

```
P4:PolynomialRing
  coordinate ring of P4
P2:PolynomialRing
  coordinate ring of P2
```

Outputs

```
X:Ideal
  of a Ionescu–Okonek surface in P4
```

Description

Text

We construct a Ionescu–Okonek surface from its rational parametrization:

```
X=P2(7;2^10,1);
```

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
P2=kk[y_0..y_2];
minimalBetti(X=ionescuOkonekSurfaceD8S5(P4,P2))
degree X, sectionalGenus X
```

```
///
```

```
///
```

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
P2=kk[y_0..y_2];
minimalBetti(X=ionescuOkonekSurfaceD8S5(P4,P2))
degree X, sectionalGenus X
X4=ideal (gens X){0..4};
minimalBetti X4
Y=X4:X;
degree Y, sectionalGenus Y
singX=singularLocus(P4/X);
dim singX
dim(X+Y)
apply(decompose(X+Y),c->(dim c, degree c, genus c, minimalBetti c))
betty tateResolutionOfSurface X
///
```

```

doc ///
Key
degree10DESSurface
(degree10DESSurface,PolynomialRing,Ring)
Headline
construct the degree 10 Decker–Ein–Schreyer surface
Usage
X=degree10DESSurface(P4,E)
Inputs
P4:PolynomialRing
coordinate ring of P4
E:Ring
dual exterior algebra
Outputs
X:Ideal
of a degree 10 Decker–Ein–Schreyer surface in P4
Description
Text
We construct the surface from a randomly chosen differential T.dd_4
of the Tate resolution of the desired ideal.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
minimalBetti(X=degree10DESSurface(P4,E))
degree X, sectionalGenus X
beti(T=tateResolutionOfSurface X)
beti(T.dd_4)
///

doc ///
Key
degree10pi8RanestadSurface
(degree10pi8RanestadSurface, PolynomialRing)
Headline
construct a degree 10 sectional genus 8 Ranestad surface
Usage
X= degree10pi8RanestadSurface P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a degree 10 sectional genus 8 Ranestad surface in P4
Description
Text
We construct the surface from a carefully chosen  $H^1_*(I_X)$  module of the ideal sheaf  $I_X$ 
with Hilbert function (2,5,3).
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
minimalBetti(X=degree10pi8RanestadSurface P4)
beti(T=tateResolutionOfSurface X)
(d,sg)=(degree X, sectionalGenus X)
LeBarzN6(d,sg,1)==6
Ksquare(d,sg,1)==-4
residual=residualInQuintics X;
dim residual, degree residual, beti residual
tally apply(primaryDecomposition residual,c->(dim c, degree c, beti c,
degree (c+X), beti saturate (c+X),
tally apply(primaryDecomposition saturate (c+X),d->(dim d, degree radical d))))
Text
There are 4 6-secant lines, 3 of them are in the plane which intersects X
in a plane quartic and three points. Hence there X contains two (-1)-lines.
The adjunction process gives the data  $L_0=\{(4, 10, 8), 2, (7, 14, 8), 1, (7, 12, 6), 0, (5, 7, 3)\}$ .
The last adjoint surface is a conic bundle in  $P_5$  with 9 singular fibers.
References
Ranestad, DES
SeeAlso
enriquesSurfaceOfDegree10

```

```

adjunctionProcessData
///

--* possibly to add to degree10pi8RanestadSurface documentation
"elapsedTime (L0,L1,L2,J)=adjunctionProcess(X);" -- 88.4042s elapsed
"L0=={(4, 10, 8), 2, (7, 14, 8), 1, (7, 12, 6), 0, (5, 7, 3)}";
"betti (fj=res J) == new BettiTally from {(0,{0},0) => 1, (1,{2},2) => 3, (1,{3},3) => 3, (2,{3},3) => 2, (2,{4},4) => 6, (3,{5},5) => 3}"
new BettiTally from {(0,{0},0) => 1, (1,{2},2) => 3, (1,{3},3) => 3, (2,{3},3) => 2, (2,{4},4) => 6, (3,{5},5) => 3}
"(degree J, sectionalGenus J)==(7,3)";
"fiber=trim ideal (random(kk^1,kk^3)*transpose fj.dd_3);";
"assert(fiber == ideal gens ring J)";
Text
It follows that the last adjoint surface  $X_3 \rightarrow B$  is a conic bundle over a  $kk$ -rational conic  $B$  subset  $P^2$ 
with  $8+4-2=9$  singular fibers. The altogether  $9+3=12$  points blown up from a minimal ruled rational surface  $Y=X_3 \text{ min} \rightarrow P^1$ 
have to lie in special position. They satisfy conditions of expected codimension  $h^0(H)*h^1(H)=10$  in the Hilbert scheme
of 12 points on  $Y$ . Thus up to projectivities we expect a  $(24-6-10=8)$ -dimensional family of such surface.
Example
"betti(N=Hom(X,P4^1/X))";
"HH^0(sheaf N) == kk^(8+24)";
Text
Another count: Up to projectivities and base changes the presentation matrix of the
 $H^1$ -module of the ideal sheaf of  $X$  depends on  $(5+1+2)-4$  parameters and the choice of the
Hilbert-Burch homomorphism on  $\dim G(2,4)=4$  further parameters.
This gives the same number 8 of parameters up to projectivities.
*_

doc ///
Key
enriquesSurfaceOfDegree10
(enriquesSurfaceOfDegree10, PolynomialRing)
Headline
construct an Enriques surface of degree 10
Usage
X= enriquesSurfaceOfDegree10 P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of an Enriques surface of degree 10 in P4
Description
Text
We construct the surface from a carefully chosen  $H^1_*(I_X)$  module of the ideal sheaf  $I_X$ 
with Hilbert function (2,5,3).
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
minimalBetti(X=enriquesSurfaceOfDegree10 P4)
beti(T=tateResolutionOfSurface X)
(d,sg)=(degree X, sectionalGenus X)
LeBarzN6(d,sg,1)==6
Ksquare(d,sg,1)==-4
HdotK(d,sg)==4
R=residualInQuintics X;
dim R,degree R
elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,1);
numList
minimalBetti Y
2*sectionalGenus Y- 2== degree Y
fourPoints=saturate L2_0;
dim fourPoints,degree fourPoints
Text
The first adjunction maps blows down 4 (-1) lines. Hence the self-intersection number of the
canonical divisor on  $Y$  is  $K_Y^2=K_X^2+4=0$ . Moreover  $H_Y.K_Y=0$ . So  $K_Y$  is numerically
trivial and  $Y$  is a minimal Enriques surface.

SeeAlso
degree10pi8RanestadSurface
adjunctionProcessData
///

```

```

/// -- experiments about the adjoint surface Y=X_1 of the Enriques surface.
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
elapsedTime tally apply(5,c->(
  X=enriquesSurfaceOfDegree10 P4;
  elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,1);
  minimalBetti Y))
_*
      0 1 2 3 4 5
o142 = Tally{total: 1 12 40 56 35 8 => 5}
      0: 1 . . . . .
      1: . 7 5 . . .
      2: . 5 35 56 35 8
*_

elapsedTime tally apply(5,c->(
  X=enriquesSurfaceOfDegree10 P4;
  elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,2);
  (numList,minimalBetti Y)))
_*
-- 101.272s elapsed

      0 1 2 3 4 5
o144 = Tally{{{(4, 10, 8), 4, (7, 14, 8), 0, (7, 14, 8)}, total: 1 11 39 56 35 8} => 5}
      0: 1 . . . . .
      1: . 7 4 . . .
      2: . 4 35 56 35 8
*_

elapsedTime tally apply(2,c->(
  X=enriquesSurfaceOfDegree10 P4;
  elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,1);
  (minimalBetti Y,minimalBetti ideal (gens Y)_{0..6}))
)
_*
o145 = Tally{(total: 1 12 40 56 35 8, total: 1 7 16 16 7 1) => 2}
      0: 1 . . . . . 0: 1 . . . . .
      1: . 7 5 . . . 1: . 7 5 . . .
      2: . 5 35 56 35 8 2: . . 11 11 . .
      3: . . . 5 7 .
      4: . . . . 1
*_

--elapsedTime tally apply(2,c->(
  X=enriquesSurfaceOfDegree10 P4;
  elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,1);
  J=ideal (gens Y)_{0..6};
  fourPoints=saturate L2_0;
  degree fourPoints, dim fourPoints, minimalBetti fourPoints
  apply(decompose fourPoints,c->(dim c, degree c))
  minimalBetti J, minimalBetti Y
  codim Y
  (dim J, degree J), (dim Y, degree Y, genera Y)
  residual=J:Y;
  --cJ=primaryDecomposition J;
  degree residual, dim residual, betti residual
  cResidual=primaryDecomposition residual;
  #cResidual
  netList apply(cResidual,c->(cY=primaryDecomposition saturate(c+Y);(dim c, degree c, betti c,
    dim(c+Y),degree (c+Y), betti saturate (c+Y),genus radical (c+Y),
    tally apply(cY,d->(dim d, degree d, genus d,betti d))
    )))
  tally apply(cResidual,c->(dim(c+fourPoints),
    degree(c+fourPoints),saturate(c+fourPoints)))
M5x5=matrix apply(cResidual_{0,2,1,4,3},c-> apply(cResidual_{0,2,1,4,3},d->
  if dim(c+d+Y)==1 then degree(c+d+Y) else 0))
det M4x4
apply(cResidual_{0..2,4},c->selfIntersectionNumber(Y,c+Y))

q=cResidual_3_4
dim ideal jacobian q
codim cResidual_3, dim cResidual_3, dim (cResidual_3+Y)

```

```

support q, support cResidual_3
(minimalBetti Y, minimalBetti ideal (gens Y)_{0..6})
)

--elapsedTime tally apply(2,c->(
X=enriquesSurfaceOfDegree10 P4;
elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,2);
J=ideal (gens Y)_{0..6};
minimalBetti J, minimalBetti Y
codim Y
(dim J, degree J), (dim Y, degree Y)
residual=J:Y;
--cJ=primaryDecomposition J;
degree residual, dim residual, betti residual
cResidual=primaryDecomposition residual;
tally apply(cResidual,c->(cY=primaryDecomposition saturate(c+Y);(dim c, degree c, betti c,
dim(c+Y),degree (c+Y), betti saturate (c+Y),genus radical (c+Y),
tally apply(cY,d->(dim d, degree d, genus d,betti d))
)))
M4x4=matrix apply(cResidual,c-> apply(cResidual,d->
if dim(c+d+Y)==1 then degree(c+d+Y) else 0))
det M4x4
apply(cResidual,c->selfIntersectionNumber(Y,c+Y))

q=cResidual_3_4
dim ideal jacobian q
codim cResidual_3, dim cResidual_3, dim (cResidual_3+Y)
support q, support cResidual_3
(minimalBetti Y, minimalBetti ideal (gens Y)_{0..6}))
)

///

/// - * surface of degree 10 sectional genus 8 and x0=1 in the intersection of the two components *-
needsPackage "NongeneralTypeSurfacesInP4"
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
m2x5=transpose matrix apply(5,i->{x_i,(2*i+1)*x_i})
m2x2=matrix{{0,0},{sum(2,i->random kk*x_i^2),sum(toList(2..4),i->random kk*x_i^2)}}
betti(m2x7=map(P4^{2:-2},m2x5|m2x2))
fm2x7=res coker m2x7
betti fm2x7
fm2x7.dd_3_{0..13}^{0..14}
fm2x7.dd_4_{0..3}^{0..13}
fm2x7.dd_2_{0..14}
betti (s1=syz fm2x7.dd_2_{1..14})
betti syz s1
betti (s2=syz(transpose (s1*random(source s1,P4^{:-6})),DegreeLimit=>0))
betti (s3=syz (transpose s2_{0..8}))
betti(s4=syz transpose (fm2x7.dd_2_{1..14}*s3))
betti (X0=trim ideal (transpose s4_{2}*fm2x7.dd_2))
minimalBetti X0
(degree X0, sectionalGenus X0)=(10,8)
betti tateResolutionOfSurface X0
elapsedTime cX0=primaryDecomposition X0;#cX0==1
betti(C=saturate(X0+minors(codim X0,jacobian X0)))
dim C,degree C
cC=primaryDecomposition C;
netList apply(cC,c->(dim c, degree c, betti c, degree radical c))
cC_0, cC_1
matrix apply(cC,c-> apply(cC,d-> if dim(c+d)==1 then degree radical (c+d) else 0))
-- => X0 has two A1 lines which intersect in the point
radical cC_2

--
radical cC_3, radical cC_4
-- each of the lines contains two pinch points.
-- => in the normalization the lines become conics
twoLines=intersect(cC_0,cC_1)
--

```

```

betti(hom=Hom(C/X0,P4^1/X0))
A=prune image homomorphism hom_{1}
-- => the normalization has sectional genus 6, H1 the hyperplane class
radical cC_2
Xaff=sub(X0,x_2=>1);
p=radical(cC_2)
modP2=trim ideal(gens Xaff%p^2)

perhapsTangentCone=(modP3=trim ideal(gens Xaff%p^3))
codim modP3, dim modP3
codim perhapsTangentCone
singPTC=trim (minors(2,jacobian perhapsTangentCone)+perhapsTangentCone)
primaryDecomposition modP3
cSingPTC=primaryDecomposition singPTC;
netList apply(cSingPTC,c->(dim c, degree c, betti c))
-- => presumably it is not the tangent cone

-- guess: the two conics intersect in the normalization in a point.
cPlaneCapX0=primaryDecomposition saturate (X0+ideal(x_1,x_0))
netList apply(cPlaneCapX0,c->(dim c, degree c, degree radical c, radical c))
dim singularLocus cPlaneCapX0_2
-- => cPlaneCapX0_2 is a smooth conic

-- try to compute a normalization
c0=primaryDecomposition (ideal(x_0+x_1)+X0)
apply(c0,c->(dim c, degree c, degree radical c))
betti (R=intersect(c0_{2,3}))
R
a=R_0,b=R_1
betti(m=matrix{{b^3,a^9}}|gens X0)
betti(sm=syz m)--,DegreeLimit=>12))
sub(sm,matrix{{0_P4,0,0,0,0}})

-- for write up
binomial(9+2,2)-6-14*3-5*1
3^14, 3^8
restart
needsPackage "NongeneralTypeSurfacesInP4"
setRandomSeed ("random")
P4=ZZ/3[x_0..x_4]
elapsedTime tally apply(10,c->(X=findRandomSmoothSchreyerSurface P4;
  <<"c = "<<c<<flush <<endl;betti X))
elapsedTime X=findRandomSmoothSchreyerSurface P4;
P4=ZZ/2[x_0..x_4]
elapsedTime tally apply(10,c->(X=findRandomSmoothSchreyerSurface P4;
  <<"c = "<<c<<flush <<endl;betti X))
P4=ZZ/5[x_0..x_4]
elapsedTime tally apply(10,c->(X=findRandomSmoothSchreyerSurface P4;
  <<"c = "<<c<<flush <<endl;betti X))
2^15
kk=ZZ/nextPrime 10^4
P4=kk[x_0..x_4]
X=vBELSurface P4;
tex minimalBetti X
///

doc ///
Key
enriquesSurfaceD11S10
(enriquesSurfaceD11S10, PolynomialRing)
Headline
get an Enriques surface of degree 11 and sectional genus 10
Usage
X= enriquesSurfaceD11S10 P4
Inputs
P4:PolynomialRing
  coordinate ring of P4 over the field ZZ/3
Outputs
X:Ideal
  of an Enriques surface of degree 11 in P4
Description

```

Text

One family of the surface found as "schreyerSurface" is actually an Enriques surface. An example is the first surface in the list of precomputed schreyer surfaces.

Example

```
kk=ZZ/3;
P4=kk[x_0..x_4];
minimalBetti(X=enriquesSurfaceD11S10(P4))
beti(T=tateResolutionOfSurface X)
(d,sg)=(degree X, sectionalGenus X)
LeBarzN6(d,sg,1)==10
Ksquare(d,sg,1)==-6
X5=ideal (gens X)_{0..4};
R=X5:X;
dim R,degree R,degree(R+X)
```

Text

There are 5 six-secant lines, hence by Le Barz formula five (-1) lines. Indeed:

Example

```
"elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,4); -- 48.7283s elapsed;";
"numList=={(4, 11, 10), 5, (9, 19, 11), 1, (10, 20, 11), 0, (10, 20, 11), 0, (10, 20, 11)}";
B=new BettiTally from {(0,{0},0) => 1, (1,{2},2) => 25, (2,{3},3) => 80,
(2,{4},4) => 10, (3,{4},4) => 80, (3,{5},5) => 112, (4,{6},6) => 350,
(5,{7},7) => 400, (6,{8},8) => 245, (7,{9},9) => 80, (8,{10},10) => 11}
"degree Y==20";
"minimalBetti Y == B";
"2*sectionalGenus Y- 2== degree Y";
```

Text

The first adjunction maps blows down 5 (-1) lines. The second a (-1) conic. The second adjoint surface X2 is a minimal Enriques surface of degree 20 in a P10.

SeeAlso

```
specialEnriquesSchreyerSurface
specificSchreyerSurface
adjunctionProcessData
tateResolutionOfSurface
LeBarzN6
Ksquare
///
```

doc ///

Key

```
degree10pi9RanestadSurface
(degree10pi9RanestadSurface, PolynomialRing,Ring)
```

Headline

construct a degree 10 sectional genus 9 Ranestad surface

Usage

```
X= degree10pi8RanestadSurface(P4,E)
```

Inputs

```
P4:PolynomialRing
coordinate ring of P4
E:Ring
dual exterior algebra
```

Outputs

```
X:Ideal
of a degree 10 sectional genus 9 Ranestad surface in P4
```

Description

Text

We construct the surface from a carefully chosen differential T.dd_4 of the Tate resolution.

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
minimalBetti(X=degree10pi9RanestadSurface(P4,E))
beti(T=tateResolutionOfSurface X)
beti(T.dd_4)
degree X, sectionalGenus X
```

```
///
```

```

doc ///
Key
nonspecialAlexanderSurface
(nonspecialAlexanderSurface, PolynomialRing)
(nonspecialAlexanderSurface, PolynomialRing,PolynomialRing)
Headline
construct a nonspecial Alexander surface of degree 9
Usage
X= nonspecialAlexanderSurface P4
X= nonspecialAlexanderSurface(P4,P2)
Inputs
P4:PolynomialRing
  coordinate ring of P4
P2:PolynomialRing
  coordinate ring of P2
Outputs
X:Ideal
  of a nonspecial Alexander surface of degree 9 in P4
Description
Text
  We construct a nonspecial Alexander surface of degree 9 from its rational parametrization,
  or what is faster from a presentation of the  $H^1_*(I_X)$  module. The dual of the (3,5,1) module has
  a special presentation which gives rise to a six secant line
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
elapsedTime minimalBetti(X=nonspecialAlexanderSurface(P4))
(L0,adjList,ptsList,J)=adjunctionProcess X;
betti(T=tateResolutionOfSurface X)
Text
  Lebarz formula computes the number of 6-secant lines + the number of (-1) lines.
Example
LeBarzN6(degree X, sectionalGenus X,1)
X5=ideal (gens X)_0..14;
L=X5:X -- L is the six secant line
degree(L+X)==6
Text
  We can obtain information about this surface
  from the adjunctionProcess.
Example
(L0,adjList,ptsList,J)=adjunctionProcess(X);
ring J
betti(H=parametrization(ring J,adjList))
cH=primaryDecomposition ideal H;
tally apply(cH,c->(dim c, degree radical c, degree c))
Text
  H is a linear system of forms of degree which vanish in 10 points with
  multiplicity 4. However over the field the 10 point split into orbits
  under Frobenius.
  In the second version of the function we start with
  a rational P2 --> P4 defined by forms
  of degree 13 which vanishes on 10 randomly chosen
  points with multiplicity 4.
Example
P2=kk[y_0..y_2];
elapsedTime minimalBetti(X=nonspecialAlexanderSurface(P4,P2))
(L0,adjList,ptsList,J)=adjunctionProcess(X);
betti(H=parametrization(ring J,adjList))
cH=primaryDecomposition ideal H;
tally apply(cH,c->(dim c, degree radical c, degree c))
Text
  This times the ideal H decomposes in to 10 points of degree 1 defined over kk
  and an embedded (y_0..y_2)-primary ideal.
SeeAlso
enriquesSurfaceOfDegree9
tateResolutionOfSurface
LeBarzN6
adjunctionProcess
///
doc ///

```

Key

enriquesSurfaceOfDegree9
(enriquesSurfaceOfDegree9, PolynomialRing)

Headline

construct a Enriques surface of degree 9

Usage

X= enriquesSurfaceOfDegree9 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of an Enriques surface of degree 9 in P4

Description

Text

We construct the Enriques surface from a presentation of the $H^1_*(I_X)$ module. The dual of the (3,5,1) module is defined by 12 quadrics and completely determines X. In the dual projective space $P4^*$ this corresponds to a canonical curve of genus 5 defined by $3=15-12$ quadrics. The Enriques surface is non-minimal. It is the projection of a Fano polarized minimal Enriques surface in $P5$. Thus the universal family of the Hilbert scheme of Fano polarized Enriques surfaces is birational to the Hilbert scheme of canonical curves of genus 5 in $P5$.

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
elapsedTime minimalBetti(X=enriquesSurfaceOfDegree9(P4))
beti(T=tateResolutionOfSurface X)
(d,sg)=(degree X, sectionalGenus X)
Ksquare(d,sg,1)=-1
LeBarzN6(d,sg,1)
elapsedTime (numList,L1,L2,Y)=adjunctionProcess(X,1);
numList
minimalBetti Y
2*sectionalGenus Y -2== degree Y
point=saturate L2_0
dim point
```

Text

The self-intersection number of the canonical divisor on the first adjoint surface Y is $K_Y^2=K_X^2+1=0$. Moreover $H_Y.K_Y=0$. Hence K_Y is numerically trivial and Y is a minimal Enriques surface.

SeeAlso

nonspecialAlexanderSurface
adjunctionProcessData

///

doc ///

Key

specialityOneAlexanderSurface
(specialityOneAlexanderSurface, PolynomialRing, Ring)

Headline

construct a speciality One AlexanderSurface Alexander surface of degree 9

Usage

X= specialityOneAlexanderSurface(P4,E)

Inputs

P4:PolynomialRing

coordinate ring of P4

P2:PolynomialRing

coordinate ring of P2

Outputs

X:Ideal

of a speciality one Alexander surface of degree 9 in P4

Description

Text

We construct a speciality 1 Alexander Surface of degree 9 from its differential $T.dd_4$ of the Tate resolution.

Example

```

kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
elapsedTime minimalBetti(X=specialityOneAlexanderSurface(P4,E))
degree X, sectionalGenus X
beti(T=tateResolutionOfSurface X)
beti(T.dd_4)
beti res(coker random(target T.dd_4,source T.dd_4),LengthLimit=>4)
beti res(coker transpose random(target T.dd_4,source T.dd_4),LengthLimit=>5)

```

Text
Thus a **random** choice of the differential **T.dd_4** leads to a surface and the component of the **Hilbert** scheme is unirational

```

///
doc ///
Key
popescuSurface
(popescuSurface, PolynomialRing,Ring,Number)
Headline
construct a Popescu surface degree 11 and sectional genus 11 (3 families)
Usage
X= popescuSurface(P4,E,s)
Inputs
P4:PolynomialRing
coordinate ring of P4
E:Ring
dual exterior algebra
s: Number
the number of desired 6-secants
Outputs
X:Ideal
of a Popescu surface in P4
Description
Text
The Popescu surfaces come in three families, distinguished by their
number of 6-secant lines.
One has to choose the differential T.dd_4 suitable.
In the first case X has no 6-secant, since the ideal is generated by quintics.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
minimalBetti(X=popescuSurface(P4,E,0))
(d,sg)=(degree X, sectionalGenus X)
beti(T=tateResolutionOfSurface X)
Text
In the second case there is a unique 6-secant line.
Example
elapsedTime minimalBetti(X=popescuSurface(P4,E,1))
L=residualInQuintics X;
degree L, degree(L+X)
elapsedTime (numList,L1,L2,J)=adjunctionProcess(X,1);
numList_1
Text
The entry numList_1 is the number of (-1) lines on X. Thus we must have
Example
LeBarzN6(d,sg,1)==6+1
Text
In the third case there is a pencil of 6-secant line. Every line in
the plane through the point is a 6-secant line,
since the plane intersects the surface in a plane quintic curve.
Example
elapsedTime minimalBetti(X=popescuSurface(P4,E,2))
R=residualInQuintics X;
tally apply(primaryDecomposition (R+X),c->(dim c,degree radical c,degree(c+R)))

```

SeeAlso
adjunctionProcessData
residualInQuintics

```

///

doc ///
Key
vBELSurface
(vBELSurface, PolynomialRing)
(vBELSurface, Ring, Ring)
Headline
construct a von Bothmer–Erdenberger–Ludwig surface, degree 11, sectional genus 11
Usage
X= vBELSurface(P4,P2)
X= vBELSurface(P4)
Inputs
P4:PolynomialRing
coordinate ring of P4
P2:PolynomialRing
coordinate ring of P2
Outputs
X:Ideal
of a vBEL surface in P4
Description
Text
The first version gives the original vBEL surface defined over a field of characteristic 2.
The second version gives a construction of a vBEL surface building on a unirational liaison
construction.
Example
P4=ZZ/2[x_0..x_4]; P2=ZZ/2[u_0..u_2];
minimalBetti(X=vBELSurface(P4,P2))
ci=ideal ((gens X)_0){(gens X)*random(source gens X,P4^{5})};
Y=ci:X;
setRandomSeed("fast decomposition")
cY=decompose Y;
tally apply(cY, c-> (dim c, degree c, minimalBetti c))
betti(T=tateResolutionOfSurface X)
Text
The linked surface consists of a plane, a quadric surface and a Bordiga surface.
The unirational construction is a reversal of this linkage.
Example
kk=ZZ/nextPrime 10^3; P4=kk[x_0..x_4];
minimalBetti(X=vBELSurface P4)
betti tateResolutionOfSurface X
(L0,L1,L2,J)=adjunctionProcess(X);
L0
X45=ideal (gens X)_0..5;
R=X45:X;
dim R, degree R;
cR=decompose R;
tally apply(cR,c->(dim c, degree c, betti c))
tally apply(cR,c->degree(c+X))
LeBarzN6(11,11,1)
Text
X has two 6-secant lines and five (-1)-lines.
SeeAlso
adjunctionProcessData

///

///
-- the (4,5) linked surface of a vBEL surface can be irreducible:
kk=ZZ/nextPrime 10^3; P4=kk[x_0..x_4];
minimalBetti(X=vBELSurface P4)
singX=saturate(X+minors(2,jacobian X));
dim singX
betti (T=tateResolutionOfSurface X)
betti(T.dd_4)_3
ci=ideal ((gens X)_0){(gens X)*random(source gens X,P4^{5})};
Y=ci:X;
--setRandomSeed("fast decomposition")
elapsedTime cY=decompose Y;-- 328.197 seconds elapsed
tally apply(cY, c-> (dim c, degree c, minimalBetti c))

```

```

singY=saturate(Y+minors(2,jacobian Y));
minimalBetti singY
dim singY, degree singY
csingY=primaryDecomposition singY;
tally apply(csingY,c->(dim c, degree c, minimalBetti c))
betti(omegaY=Ext^1(module Y,P4^{-5}))
betti (m4xx=syz (m6x13=presentation omegaY)^{4,5})
betti(m4xr=m6x13^{0..3}*m4xx)
minimalBetti coker m4xr
P3=kk[y_0..y_3]
P3xP4=P3**P4
betti(graph=ideal(sub(vars P3,P3xP4)*sub(m4xr,P3xP4)))
betti (sg=saturate(graph,sub(ideal gens P4,P3xP4)))
minimalBetti(inP3=trim sub(sg,P3))
cinP3=decompose(inP3)
tally apply(cinP3,c->(dim c, degree c))
quadric=first cinP3
while(
  line=ideal random(P3^1,P3^{2:-1});
  pts=decompose (line + quadric);
  #pts<2) do ()
pt=sub(transpose syz transpose jacobian first pts,kk)
fib=trim ideal(pt*m4xr)
fib0=trim ideal(random(kk^1,kk^4)*m4xr)
m4x4=m4xr_{0..3}
ann coker m4x4==ideal Y_0
singY0=saturate ideal jacobian Y_0;
dim singY0, degree singY0
apply(decompose singY0,c->(dim c, degree c, minimalBetti c))
line1=(decompose singY0)_1
line=(decompose singY0)_0
betti (H=intersect(line,line1))
cH=primaryDecomposition saturate(Y+ideal H_0);
tally apply(cH,c->(dim c, degree c, minimalBetti c, minimalBetti radical c, genus c))
degree(line+X)
apply(5,i->(dim(line^i+Y),degree(line^i+Y)))
degree Y,sectionalGenus Y
intersect
///
doc ///
Key
randomSurfaceDegreeAndSectionalGenus
(randomSurfaceDegreeAndSectionalGenus, Function, List)
Headline
construct of a random rational surface of a given type
Usage
dsg = randomSurfaceDegreeAndSectionalGenus(F,ringList)
Inputs
F:Function
  which produces a surface in P4
ringList:List
  {P4,E,P2} of the coordinate ring of P4, the corresponding exterior algebra and the coordinate ring of P2
Outputs
dsg:List
  of the degree d and the sectional genus of the surface
Description
Text
The unirational components of the Hilbert scheme of smooth surfaces in P4 uses in the construction
P4, or P4 and E, or P4 and P2. This functions calls the corresponding function of the construction
The function computes examples of the desired surface and prints
some numerical information. It returns the degree and the sectional genus
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
P2=kk[y_0..y_2];
ringList={P4,E,P2};
Fs=
{bordigaSurface,ionescuOkonekSurfaceD7,okonekSurfaceD8S6,ionescuOkonekSurfaceD8S5,specialityOneAlexanderSurface}
elapsedTime dges=apply(Fs,F->randomSurfaceDegreeAndSectionalGenus(F,ringList))
Fs1={degree10pi8RanestadSurface,degree10pi9RanestadSurface,degree10DESSurface};

```

```

--apply(Fs1,F->elapsedTime randomSurfaceDegreeAndSectionalGenus(F,ringList))
Text
The last command should not be executed since it takes too much time.

Caveat

SeeAlso

///

doc ///
Key
quinticEllipticScroll
(quinticEllipticScroll, PolynomialRing)
Headline
construct a quintic elliptic scroll
Usage
X=quinticEllipticScroll P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a quintic elliptic scroll
Description
Text
We construct an quintic elliptic scroll, which is a nonsingular surface of
degree 5 and sectional genus 1. It is linked via two cubics to the Veronese surface
of degree 4.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=quinticEllipticScroll P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
beti(T=tateResolutionOfSurface X)
ci=ideal(gens X* random(source gens X,P4^{2:-3}));
Y=ci:X; -- Y is a Veronese surface
minimalBetti Y
(degree Y, sectionalGenus Y)=(4,0)
SeeAlso
veroneseSurface
tateResolutionOfSurface
///

doc ///
Key
ellipticConicBundle
(ellipticConicBundle, PolynomialRing)
Headline
construct a elliptic conic bundle
Usage
X=ellipticConicBundle P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a elliptic conic bundle
Description
Text
We construct an elliptic conic bundle, which is a nonsingular surface of
degree 8 and sectional genus 5.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=ellipticConicBundle P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
beti(T=tateResolutionOfSurface X)

```

Text

These surface were overlooked by Okonek and Ionescu respectively in their classification of low degree smooth projective surfaces.

References

Okonek
Ionescu
Abo et al

SeeAlso

tateResolutionOfSurface
///

doc ///

Key

biellipticSurfaceD10
(biellipticSurfaceD10,PolynomialRing)

Headline

construct a bielliptic surface of degree 10

Usage

X=biellipticSurfaceD10 P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal
of a bielliptic surface of degree 10

Description

Text

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];  
elapsedTime X=biellipticSurfaceD10 P4;  
(d,sg)=(degree X, sectionalGenus X)  
minimalBetti X  
betti(T=tateResolutionOfSurface X)
```

Text

The construction uses Moore matrices and a search for 6 torsions point on an elliptic curve.
Perhaps the code can be improved.

References

ADHPR

SeeAlso

tateResolutionOfSurface
///

doc ///

Key

biellipticSurfaceD15
(biellipticSurfaceD15,PolynomialRing)

Headline

construct a bielliptic surface of degree 15

Usage

X=biellipticSurfaceD10 P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal
of a bielliptic surface of degree 10

Description

Text

Example

```
kk=ZZ/nextPrime 10^4;  
P4=kk[x_0..x_4];  
elapsedTime X=biellipticSurfaceD15 P4;  
(d,sg)=(degree X, sectionalGenus X)  
minimalBetti X  
betti(T=tateResolutionOfSurface X)
```

Text

The construction uses Moore matrices and a search for 6 torsions point on an elliptic curve .
Perhaps the code can be improved.

Caveat

The function can fail, in which case it returns null.

References

ADHPR

SeeAlso

tateResolutionOfSurface

///

doc ///

Key

abelianSurfaceD10

horrocksMumfordSurface

(horrocksMumfordSurface,PolynomialRing)

(abelianSurfaceD10,PolynomialRing)

Headline

construct a nonsingular abelian surface of degree 10 and sectional genus 6

Usage

X=abelianSurfaceD10 P4

X=horrocksMumfordSurface P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of a abelian surface of degree 10, a Horrocks–Mumford surface.

Description

Text

Horrocks and Mumford rediscovered these surfaces as the zero locus of sections of the Horrocks–Mumford bundle, which is a rank 2 vector bundle with

Chern classes $c_1=-1$ and $c_2=4$.

Comesati found these surfaces already in 192x?

Example

```
kk=ZZ/nextPrime 10^4;
```

```
P4=kk[x_0..x_4];
```

```
X=abelianSurfaceD10 P4;
```

```
betti(fX=res X)
```

```
(d,sg)=(degree X, sectionalGenus X)
```

```
betti(T=tateResolutionOfSurface(X,7))
```

```
k2=Ksquare(10,6,0)
```

```
LeBarzN6(10,6,0)==25
```

```
tally apply(decompose residualInQuintics(X),c->(dim c-1, degree c,  
(dim(c+X)-1 ,degree(c+X))))
```

Text

X has 25 six-secant lines. It is linked in a (5,5) complete intersection to a surface X' of degree 15. The six secant lines are twentyfive (-1) -lines of X' .

Example

```
ci=ideal(gens X*random(source gens X,P4^{2:-5}));
```

```
X'=ci:X;
```

```
minimalBetti X'
```

```
betti tateResolutionOfSurface(X',7)
```

```
(d',sg')=(degree X',sectionalGenus X')
```

```
Ksquare(d',sg',0)==-25
```

```
LeBarzN6(d',sg',0)==LeBarzN6(d,sg,0)
```

```
saturate residualInQuintics(X')
```

Text

The surface X is the zero loci of a vector bundle, whose

module of global sections is

Example

```
HMBundle=coker transpose (syz transpose fX.dd_3)_{0..18};
```

```
minimalBetti HMBundle
```

Text

In the code we use the Horrocks–Mumford bundle to get X.

The construction of the Horrocks–Mumford bundle uses a monad:

Example

```
e=symbol e;
```

```
E=kk[e_0..e_4,SkewCommutative=>true];
```

```
alphanad= map(E^{5:0},E^{2:-2},{e_4*e_1,e_2*e_3},{e_0*e_2,e_3*e_4},{e_1*e_3,e_4*e_0},{e_2*e_4,e_0*e_1},  
{e_3*e_0,e_1*e_2});
```

Text

The matrix

Example

```
diagonalMatrix{1,-1}*transpose alphanad
```

Text

is the famous Horrocks–Mumford [matrix](#) which leads to a Tate [resolution](#) of the following shape

Example

```
F=res coker alphad;  
betti (F=res(coker transpose F.dd_6,LengthLimit=>10)[5]**E^{0})
```

Text

The Horrocks–Bundle is obtained as the [homology](#) of a monad. The [module](#) HMBundle below is the [module](#) of [global](#) sections of the Horrocks–Mumford bundle.

Example

```
HMBundle=(prune homology(beilinson(F'.dd_0**E^{-4},P4),beilinson(F'.dd_1**E^{-4},P4)))*P4^{-4}  
minimalBetti HMBundle  
minimalBetti X
```

Text

The [Hilbert](#) function of the [cohomology](#) of the Horrocks–Mumford bundle is incoded in the Tate [resolution](#), cf. [EFS,pp xxx].

Example

```
H2cohomology=prune Ext^2(HMBundle,P4^{-5})  
H1cohomology=prune Ext^1(HMBundle,P4^{-5})  
apply(toList(1..6),i->hilbertFunction(i,H1cohomology))  
betti F'
```

References

Horrocks–Mumford
Barth–Hulek–Moore
Comessati
Decker–Schreyer
Eisenbud–Floyd–Schreyer

SeeAlso

///

doc ///

Key

abelianSurfaceD15
(abelianSurfaceD15, [PolynomialRing](#))

Headline

construct an abelian surface of degree 15

Usage

X=abelianSurfaceD15 P4

Inputs

P4:[PolynomialRing](#)
coordinate ring of P4

Outputs

X:[Ideal](#)
of an abelian surface of degree 15

Description

Text

These abelian surfaces are linked via two quintics to a Horrocks–Mumford surface. The construction uses this liaison.

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];  
X=abelianSurfaceD15 P4;  
(d,sg)=(degree X, sectionalGenus X)  
minimalBetti X  
betti (T=tateResolutionOfSurface(X,7))  
HdotK(d,sg)==25  
xO=chiO(X)  
Ksquare(d,sg,xO)==-25
```

Text

X is a non-minimal abelian surface. it contains twentyfive (–1) [lines](#).
X is linked via two quintics to a Horrocks–Mumford surface Y.

Example

```
ci=ideal(gens X* random(source gens X,P4^{2:-5}));  
Y=ci:X;  
minimalBetti Y  
(degree Y, sectionalGenus Y) == (10,6)  
betti tateResolutionOfSurface(Y,7)
```

References

Horrocks–Mumford
BHM

SeeAlso

horrocksMumfordSurface

```

adjunctionProcessData
///
--*
Text
The surface X is non-minimal. Adjunction gives
Example
betti (HplusK=presentation truncate(1,Ext^1(X,P4^{5})))
P19=kk[y_0..y_19];
P4xP19=P4**P19;
graph=sub(vars P19,P4xP19)*sub(HplusK,P4xP19);
betti(presH=map(P19^5,,sub(contract(transpose sub(vars P4,P4xP19),graph),P19)))
elapsedTime betti(X1=ann coker presH) -- 49.985s elapsed
minimalBetti X1
Text
X1 is the first adjoint surface

Example
(dim X1,degree X1,sectionalGenus X1)==(3,40,21)
Text
The image H1 of a general hyperplane section of X under the adjunction map is a
canonical curve, however projected since  $h^2(\omega_X)=1$ .
Example
elapsedTime H1=ann coker(random(P19^4,P19^5)*presH)
beti H1
(dim H1, degree H1, genus H1)==(2,40,21)
elapsedTime pts=sum apply(3,i->ann coker(random(P19^4,P19^5)*presH));
(dim pts, degree pts)==(1,25)
Text
the ideal pts is the ideal of the intersection the of three general hypersurface.
Thus this ideal defines the exceptional points of the first adjunction map  $X \rightarrow X1$ .
Indeed their preimage coincides with the 25 Horrocks-Mumford lines, see [HM]
The structure sheaf of  $\mathcal{O}_X$  has Euler characteristic 0 and  $K^2=-25$ 
Example
 $\mathcal{O}_X=\text{sheaf}(P4^1/X)$ ;
apply(3,i->HH^i( $\mathcal{O}_X$ ))
Ksquare(degree X, sectionalGenus X, 0)==-25
Text
Thus the surface X1 is minimal.
*_
doc ///
Key
abelianSurfaceD15S21Popescu
(abelianSurfaceD15S21Popescu, PolynomialRing)
Headline
construct an abelian surface of degree 15 and sectional genus 21
Usage
X=abelianSurfaceD15S21Popescu P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of an abelian surface of degree 15 and sectional genus 21
Description
Text
These surfaces are not linked to Horrocks-Mumford surfaces. The construction builds upon a subtle
use of three Moore matrices.
Example
kk=ZZ/nextPrime 10^4;
P4=kk[x_0..x_4];
X=abelianSurfaceD15S21Popescu P4;
dim singularLocus X == 0 ==> X is smooth
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
beti tateResolutionOfSurface(X,7)
HdotK(d,sg)==25
xO=chiO(X)
Ksquare(d,sg,xO)==-25
LeBarzN6(d,sg,xO)
Text
X is a non-minimal abelian surface. It contains twenty five (-1) lines.

```

References

Popescu

SeeAlso

horrocksMumfordSurface

abelianSurfaceD15

///

_*

```
ci=ideal(gens X*random(source gens X,P4^{ -5,-6}));
```

```
Y=ci:X;
```

```
minimalBetti Y
```

```
d=degree Y, sg=sectionalGenus Y
```

```
xO=chiO(Y)
```

```
Ksquare(d,sg,xO)
```

```
betti tateResolutionOfSurface(Y,8)
```

*_

doc ///

Key

ellipticSurfaceD7

(ellipticSurfaceD7,PolynomialRing)

Headline

construct an elliptic surface of degree 7

Usage

X=ellipticSurfaceD7 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of an elliptic surface of degree 7

Description

Text

Example

```
kk=ZZ/nextPrime 10^3;
```

```
P4=kk[x_0..x_4];
```

```
elapsedTime X=ellipticSurfaceD7 P4;
```

```
(d,sg)=(degree X, sectionalGenus X)
```

```
minimalBetti X
```

```
betti(T=tateResolutionOfSurface X)
```

Text

The surface is ACM. We use its Hilbert–Burch matrix.

References

SeeAlso

tateResolutionOfSurface

///

doc ///

Key

ellipticSurfaceD8

(ellipticSurfaceD8,PolynomialRing)

Headline

construct an elliptic surface of degree 8

Usage

X=ellipticSurfaceD8 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of an elliptic surface of degree 8

Description

Text

Example

```
kk=ZZ/nextPrime 10^3;
```

```
P4=kk[x_0..x_4];
```

```
elapsedTime X=ellipticSurfaceD8 P4;
```

```
(d,sg)=(degree X, sectionalGenus X)
```

```
minimalBetti X
```

```
betti(T=tateResolutionOfSurface X)
```

```
pg=rank HH^2(sheaf(P4/X))
```

Text

The surface is ACM. We use its [Hilbert–Burch matrix](#).
[References](#)

[SeeAlso](#)

tateResolutionOfSurface
///

doc ///

Key

ellipticSurfaceD9
(ellipticSurfaceD9,PolynomialRing)

Headline

construct an elliptic surface of degree 9

Usage

X=ellipticSurfaceD9 P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal
of an elliptic surface of degree 9

Description

Text

The syzygy bundle G of the desired H^1 -module has rank 5
and a presentation 7×15 matrix

Example

new BettiTally from {(0,{2},2) => 2, (1,{3},3) => 7, (2,{4},4) => 5, (2,{5},5) =>
10}

Text

We take the dependency locus of a homomorphism from $F=3O(-4)+O(-5)$ to G .

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];  
elapsedTime X=ellipticSurfaceD9 P4;  
(degree X, sectionalGenus X)=(9,7)  
minimalBetti X  
betty(T=tateResolutionOfSurface X)  
pg=rank HH^2(sheaf(P4/X))  
Ksquare(9,7,2)==0  
D=saturate canonicalDivisor X;  
betty D, degree D  
HdotK(9,7)==degree D  
selfIntersectionNumber(X,D)
```

Text

X is a minimal elliptic surface.

[References](#)

[SeeAlso](#)

tateResolutionOfSurface
canonicalDivisor
selfIntersectionNumber
Ksquare
///

doc ///

Key

ellipticSurfaceD10S9
(ellipticSurfaceD10S9,PolynomialRing)

Headline

construct an elliptic surface of degree 10 and sectional genus 9

Usage

X=ellipticSurfaceD10S9 P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal
of an elliptic surface of degree 10 and sectional genus 9

Description

Text

Example

```

kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
setRandomSeed(" fix the decomposition of the canonicalDivisor");
elapsedTime X=ellipticSurfaceD10S9 P4;
(degree X, sectionalGenus X)=(10,9)
minimalBetti X
betty(T=tateResolutionOfSurface X)
pg=rank HH^2(sheaf(P4/X))
LeBarzN6(10,9,2)
HdotK(10,9)
Ksquare(10,9,2)
D=saturate canonicalDivisor X;
apply(primaryDecomposition D,c->(dim c,degree c,genus c,selfIntersectionNumber(X,c)))

```

Text

X is non-minimal elliptic surface blown-up in three points.

References

SeeAlso

tateResolutionOfSurface
canonicalDivisor
selfIntersectionNumber
Ksquare
LeBarzN6

///

doc ///

Key

ellipticSurfaceD10S10

(ellipticSurfaceD10S10,PolynomialRing)

Headline

construct an elliptic surface of degree 10 and sectional genus 10

Usage

X=ellipticSurfaceD10S10 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of an elliptic surface of degree 10 and sectional genus 10

Description

Text

Example

```

kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
elapsedTime X=ellipticSurfaceD10S10 P4;
(degree X, sectionalGenus X)=(10,10)
minimalBetti X
betty(T=tateResolutionOfSurface X)
pg=rank HH^2(sheaf(P4/X))
LeBarzN6(10,10,3)

D=saturate canonicalDivisor X;
tally apply(primaryDecomposition D,c->
(dim c,degree c,genus c,minimalBetti c,selfIntersectionNumber(X,c)))
HdotK(10,10)==2+6
Ksquare(10,10,3)==-2

```

Text

X is elliptic surface blown-up with two (-1) lines.

References

SeeAlso

tateResolutionOfSurface
canonicalDivisor
HdotK
Ksquare
LeBarzN6

///

doc ///

Key

ellipticSurfaceD11S12

(ellipticSurfaceD11S12,PolynomialRing)

[Headline](#)

construct an elliptic surface of degree 11 and sectional genus 12

[Usage](#)

X=ellipticSurfaceD11S12 P4

[Inputs](#)

P4:PolynomialRing

coordinate ring of P4

[Outputs](#)

X:Ideal

of an elliptic surface of degree 11 and sectional genus 12

[Description](#)

[Text](#)

[Example](#)

kk=ZZ/nextPrime 10^3;

P4=kk[x_0..x_4];

elapsedTime X=ellipticSurfaceD11S12 P4;

(degree X, sectionalGenus X)=(11,12)

minimalBetti X

betti(T=tateResolutionOfSurface X)

pg=rank HH^2(sheaf(P4/X))

D=saturate canonicalDivisor X;

tally apply(primaryDecomposition D,c->

(dim c,degree c,genus c,minimalBetti c,selfIntersectionNumber(X,c)))

LeBarzN6(11,12,3)==3

HdotK(11,12)==3+6+2

Ksquare(11,12,3)

[Text](#)

X is an elliptic surface blown-up with three (-1) lines and one (-1) conic.

[References](#)

[SeeAlso](#)

tateResolutionOfSurface

canonicalDivisor

HdotK

Ksquare

LeBarzN6

///

doc ///

[Key](#)

ellipticSurfaceD12S14L0

(ellipticSurfaceD12S14L0,PolynomialRing)

[Headline](#)

construct an elliptic surface of degree 12, sectional genus 14 and no 6-secant line

[Usage](#)

X=ellipticSurfaceD12S14L0 P4

[Inputs](#)

P4:PolynomialRing

coordinate ring of P4

[Outputs](#)

X:Ideal

of an elliptic surface of degree 12 and sectional genus 14

[Description](#)

[Text](#)

[Example](#)

kk=ZZ/nextPrime 10^3;

P4=kk[x_0..x_4];

elapsedTime X=ellipticSurfaceD12S14L0 P4;

(degree X, sectionalGenus X)=(12,14)

minimalBetti X

betti(T=tateResolutionOfSurface X)

pg=rank HH^2(sheaf(P4/X))

D=saturate canonicalDivisor X;

degree D,HdotK(12,14)

selfIntersectionNumber(X,D),Ksquare(12,14,3)

[Text](#)

X is a non-minimal elliptic surface with four (-1) lines and one further (-1) curve.

[References](#)

[SeeAlso](#)

```

tateResolutionOfSurface
canonicalDivisor
selfIntersectionNumber
Ksquare
LeBarzN6
///

doc ///
Key
ellipticSurfaceD12S14Linfinite
(ellipticSurfaceD12S14Linfinite,PolynomialRing)
Headline
construct an elliptic surface of degree 12, sectional genus 14 and infinitely many 6-secant line
Usage
X=ellipticSurfaceD12S12Linfinite P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of an elliptic surface of degree 12 and sectional genus 14
Description
Text
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
elapsedTime X=ellipticSurfaceD12S14Linfinite P4;
(degree X, sectionalGenus X)=(12,14)
minimalBetti X
beti(T=tateResolutionOfSurface X)
LeBarzN6(12,14,3)
HdotK(12,14)
Ksquare(12,14,3)
R=residualInQuintics X;
dim R, degree R ,beti R, degree (R+X)
apply(primaryDecomposition(R+X),c->(dim c ,degree c, beti c))
Text
X contains a plane quintic curve with an additional point p in that plane.
Every line through p in that plane is a 6-secant line.
References

SeeAlso
tateResolutionOfSurface
residualInQuintics
Ksquare
LeBarzN6
///

doc ///
Key
ellipticSurfaceD12S13
(ellipticSurfaceD12S13,PolynomialRing)
Headline
construct a regular elliptic surface of degree 12 and sectional genus 13
Usage
X=ellipticSurfaceD12S13 P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a regular elliptic surface of degree 12 and sectional genus 13
Description
Text
We construct a regular elliptic surface of degree 12, sectional genus 13 and pg=2.
Example
kk=ZZ/nextPrime 10^4;
P4=kk[x_0..x_4];
X=ellipticSurfaceD12S13 P4;
(d,sg)=(degree X, sectionalGenus X)

```

```

minimalBetti X
beti(T=tateResolutionOfSurface X)
Ksquare(d,sg,3)
D=saturate canonicalDivisor X;
HdotK(d,sg)
degree D
rank HH^0 sheaf(P4^1/D)
selfIntersectionNumber(X,D)
genus D
singD=singularLocus(P4/D); dim singD==0 -- 9.57484s elapsed
minimalBetti D

```

Text

Since $K^2=0$ and the canonical divisor connected this surface is minimal. The surface is elliptic fibration over P^1 into elliptic curves of degree 12. Perhaps use regular in codimension 1 from FastMinors

References

SeeAlso

```

tateResolutionOfSurface
selfIntersectionNumber
canonicalDivisor
HdotK
Ksquare
///

```

doc ///

Key

```

specificEllipticSurfaceD13S16
(specificEllipticSurfaceD13S16,PolynomialRing, Ring, ZZ)

```

Headline

construct a specific elliptic surface via linkage form a Abo surface (3 families)

Usage

```

X=specificEllipticSurfaceD13S16(P4,E,k)

```

Inputs

P4:PolynomialRing

coordinate ring of P4

E: Ring

exterior algebra dual to P4

k: ZZ

the number of the specific Abo surface used in the construction.

Outputs

X:Ideal

of a specific elliptic surface of degree 13 sectional genus 16

Description

Text

We construct a specific elliptic surface of degree 13 sectional genus 16 from a specific AboSurface of number k via linkage. The function needs the ground field $kk=ZZ/19$ and a number k in {1,2,4}.

Example

```

kk=ZZ/19;
P4=kk[x_0..x_4];
E=kk[e_0..e_4,SkewCommutative=>true];
setRandomSeed("faily fast");
X=specificEllipticSurfaceD13S16(P4,E,2);

```

Text

The printed information is the partition type of canonical divisor of the specific Abo surface used and its residual in Quintics decomposition. Which in this case consists of four 6-secant lines and a rational normal curve of degree 4 which is 21-secant.

Example

```

minimalBetti X
beti(T=tateResolutionOfSurface(X))
(d,sg)=(degree X, sectionalGenus X)
Ksquare(d,sg,3)
HdotK(d,sg)
K1=canonicalDivisor X;
K2=canonicalDivisor X;
beti(baseLocus=saturate (K1+K2))
cBaseLocus=decompose baseLocus;
minimalBetti last cBaseLocus
tally apply(cBaseLocus,c->(dim c, degree c, genus c, selfIntersectionNumber(X,c)))

```

Text

Note that the exceptional curves of X coincide with the residual quintics curves of the used Abo surface.

Example

```
E1=K1:baseLocus;  
dim E1, degree E1, genus E1, selfIntersectionNumber(X,E1)  
R=residualInQuintics X;  
tally apply(decompose R,c->(dim c, degree c, genus c, degree (c+X)))
```

Text

X contains four (-1) lines and a (-1) degree 4 rational normal curve. The canonical divisor E of the minimal surface has genus 1 and self-intersection number 0. Thus X is an elliptic surface.

References

SeeAlso

tateResolutionOfSurface
selfIntersectionNumber
canonicalDivisor
HdotK
Ksquare

///

///

///

_*

regularInCodimension(1,P4/D)

*_

doc ///

Key

irregularEllipticSurfaceD12

(irregularEllipticSurfaceD12,PolynomialRing)

Headline

construct a regular elliptic surface of degree 12 and sectional genus 13

Usage

X=irregularEllipticSurfaceD12 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of a irregular elliptic surface of degree 12 sectional genus 13

Description

Text

We construct a irregular elliptic surface of degree 12 sectional genus 13 and $pg=3$.

Example

```
kk=ZZ/nextPrime 10^4;  
P4=kk[x_0..x_4];  
setRandomSeed("fix a fast canonical divisor");  
X=irregularEllipticSurfaceD12 P4;  
(d,sg)=(degree X, sectionalGenus X)  
minimalBetti X  
beti(T=tateResolutionOfSurface(X,7))  
Ksquare(d,sg,3)  
"D=saturate canonicalDivisor X;";  
HdotK(d,sg)  
"degree D==HdotK(d,sg);"  
"beti(fD=res D);"  
"M=(coker transpose fD.dd_4)**P4^{5}";  
"hilbertFunction(0,M)+1==3 -- the number of connected components of D";  
"selfIntersectionNumber(X,D)==0";
```

Text

Since $K^2=0$ this surface is minimal. X is fibered in elliptic curves of degree $4=12/3$

The canonical divisor is the pullback of a divisor of degree 3 on the albanese curve, which is an elliptic curve. This fits with $pg=3$.

References

SeeAlso

tateResolutionOfSurface
selfIntersectionNumber

```

canonicalDivisor
HdotK
Ksquare
///

/// -- the linked surface
beti(ci=ideal (gens X*random(source gens X,P4^{2:-5})))
minimalBetti(Y=ci:X)
beti(tateResolutionOfSurface(Y,7))
singY=saturate ideal singularLocus(P4/Y)
dim singY, degree singY, genus singY, minimalBetti singY
csingY=decompose singY
netList apply(csingY,c-->(dim c, degree c, genus c, minimalBetti c))
_*
+-----+
|          0 1 2 3 |
o436 = |(2, 1, 0, total: 1 3 3 1) |
|          0: 1 3 3 1 |
+-----+
|          0 1 2 3 4 |
|(2, 3, -2, total: 1 6 11 8 2)|
|          0: 1 1 . . . |
|          1: . 1 1 . . |
|          2: . 4 10 8 2 |
+-----+
|          0 1 2 3 4 |
|(2, 3, -2, total: 1 6 11 8 2)|
|          0: 1 1 . . . |
|          1: . 1 1 . . |
|          2: . 4 10 8 2 |
+-----+
*_
///

```

```

/// -- residual to X=irregularEllipticSurfaceD12 P4
kk=ZZ/nextPrime 10^4;
P4=kk[x_0..x_4];
setRandomSeed("fix a fast canonical divisor");
X=irregularEllipticSurfaceD12 P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
beti(T=tateResolutionOfSurface(X,7))
Ksquare(d,sg,3)
ci=ideal(gens X*random(source gens X,P4^{2:-5}));
Y=ci:X;
dim Y, degree Y, sectionalGenus Y
minimalBetti Y
beti tateResolutionOfSurface(Y,8)
Ksquare(13,16,2)
cY=decompose Y
netList apply(cY,c-->(dim c,degree c, sectionalGenus c))
Y1=cY_1;
dim singularLocus(P4/Y1)
beti tateResolutionOfSurface Y1
degree Y1,sectionalGenus Y1
Ksquare(10,10,4)
HdotK(10,10)
D1=saturate canonicalDivisor Y1;
D2=saturate canonicalDivisor Y1;
dim D1, degree D1, genus D1
tally apply(primaryDecomposition (D1+D2),c-->(dim c,degree c))
selfIntersectionNumber(Y1,D1)
-- => Y1 is a minimal surface of general type
///
_*
rank HH^0((sheaf(P4^1/D)))
cD=primaryDecomposition D; -- 5.2895s elapsed
cD=select(cD,c--> dim c ==2);
#cD

```

```

tally apply(cD,c->(dim c,degree c,degree radical c,minimalBetti c))
matrix apply(cD,c->apply(cD,d->dim (c+d)))
netList apply(cD,c->(singC=singularLocus(P4/c);
(rank HH^0 sheaf(P4^1/c),genus c,degree c,selfIntersectionNumber(X,c),dim singC)))

```

*_

```

doc ///
Key
K3surfaceD6
(K3surfaceD6, PolynomialRing)
Headline
construct a K3 surface of degree 6
Usage
X=K3surface6 P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a K3 surface of degree 6
Description
Text
We construct an K3 surface of degree 6 as a complete intersection of type (2,3).
X is minimal and has sectional genus 4.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=K3surfaceD6 P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
oX=chiO(X)
Ksquare(d,sg,oX)
beti tateResolutionOfSurface X
Text
X is nonminimal with one exceptional line.
References

SeeAlso

///

```

```

doc ///
Key
K3surfaceD7
(K3surfaceD7, PolynomialRing)
Headline
construct a K3 surface of degree 7
Usage
X=K3surfaceD7 P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a K3 surface of degree 7
Description
Text
We construct an K3 surface of degree 7 and sectional genus 5.
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=K3surfaceD7 P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
OX=sheaf(P4^1/X);
apply(3,i->HH^i(OX))
Ksquare(degree X,sectionalGenus X,2)
LeBarzN6(7,5,2)
Text

```

X is nonminimal with one exceptional line.

References

Okonek
Ionescu

SeeAlso

///

doc ///

Key

K3surfaceD8

(K3surfaceD8, PolynomialRing)

Headline

construct a K3 surface of degree 8

Usage

X=K3surfaceD8 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of a K3 surface of degree 8

Description

Text

We construct an K3 surface of degree 8 and sectional genus 6.

Example

kk=ZZ/nextPrime 10^3;

P4=kk[x_0..x_4];

X=K3surfaceD8 P4;

(d,sg)=(degree X, sectionalGenus X)

minimalBetti X

OX=sheaf(P4^1/X);

apply(3,i->HH^i(OX))

Ksquare(degree X,sectionalGenus X,2)

LeBarzN6(8,6,2)

HdotK(8,6)

Text

X is non-minimal with one exceptional curve of degree 2.

References

Okonek
Ionescu

SeeAlso

///

doc ///

Key

K3surfaceD9

(K3surfaceD9, PolynomialRing)

Headline

construct a K3 surface of degree 8

Usage

X=K3surfaceD9 P4

Inputs

P4:PolynomialRing

coordinate ring of P4

Outputs

X:Ideal

of a K3 surface of degree 9

Description

Text

We construct an K3 surface of degree 9 and sectional genus 8.

Example

kk=ZZ/nextPrime 10^3;

P4=kk[x_0..x_4];

X=K3surfaceD9 P4;

(d,sg)=(degree X, sectionalGenus X)

minimalBetti X

```

OX=sheaf(P4^1/X);
apply(3,i->HH^i(OX))
Ksquare(degree X,sectionalGenus X,2)
LeBarzN6(9,8,2)
HdotK(9,8)==5
Text
X is non-minimal with five exceptional lines.
References
Okonek
Ionescu

SeeAlso

///

doc ///
Key
K3surfaceD10S9L1
(K3surfaceD10S9L1, PolynomialRing)
Headline
construct a K3 surface of degree 10 and sectional genus 9 with 1 six-secant line
Usage
X=K3surfaceD10S9L1 P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a K3 surface of degree 9
Description
Text
We construct an K3 surface of degree 10 and sectional genus 9 with one 6-secant line
Example
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=K3surfaceD10S9L1 P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
OX=sheaf(P4^1/X);
apply(3,i->HH^i(OX))
Ksquare(degree X,sectionalGenus X,2)
LeBarzN6(10,9,2)
HdotK(10,9)
Text
X is non-minimal with two exceptional lines and three exceptional conics.
References
Okonek
Ionescu

SeeAlso

///

doc ///
Key
K3surfaceD10S9L3
(K3surfaceD10S9L3, PolynomialRing)
Headline
construct a K3 surface of degree 10 and sectional genus 9 with 3 six-secant line
Usage
X=K3surfaceD10S9L3 P4
Inputs
P4:PolynomialRing
coordinate ring of P4
Outputs
X:Ideal
of a K3 surface of degree 9
Description
Text
We construct an K3 surface of degree 10 and sectional genus 9

```

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=K3surfaceD10S9L3 P4;
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
OX=sheaf(P4^1/X);
apply(3,i->HH^i(OX))
Ksquare(degree X,sectionalGenus X,2)
LeBarzN6(10,9,2)
HdotK(10,9)
```

Text

X is non-minimal with no exceptional line and three exceptional conics.
The three 6-secant lines arise as follows:

Example

```
beti(X5=ideal (gens X)_{0..5})
beti(plane=X5:X)
dim plane == 3
dim (plane+X), degree radical (plane+X)
tally apply(primaryDecomposition (plane+X),c->(dim c, degree radical c))
```

Text

The plane intersects X in a quartic curve and three points. The three lines through two of these points are the three 6-secant lines.

References

SeeAlso

///

doc ///

Key

K3surfaceD11S11Ln
(K3surfaceD11S11Ln, PolynomialRing,ZZ)

Headline

construct a K3 surface of degree 11, sectional genus 11 and precisely n six-secants lines (4 families)

Usage

X=K3surfaceD11S11Ln(P4,n)

Inputs

P4:PolynomialRing
coordinate ring of P4
n: ZZ

the number of desired six secant lines

Outputs

X:Ideal

of a K3 surface of degree 11, sectional genus 11 and precisely 'n' six-secants lines

Description

Text

We construct an K3 surfaces of degree 11, sectional genus 11 and n six-secants lines

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
X=K3surfaceD11S11Ln(P4,0);
(d,sg)=(degree X, sectionalGenus X)
minimalBetti X
--OX=sheaf(P4^1/X);
--apply(3,i->HH^i(OX))
```

Text

X has no 6-secant line, since the ideal is generated by quintics.

Example

```
X=K3surfaceD11S11Ln(P4,1);
minimalBetti X
```

Text

In case n=1 there is precisely one 6-secant line:

Example

```
beti(X5=ideal (gens X)_{0..8})
beti(line=X5:X)
dim line, degree line, degree (line+X)
```

Text

In case n=2 there are 2 six secant lines:

Example

```
X=K3surfaceD11S11Ln(P4,2);
minimalBetti X
betti(X5=ideal (gens X)_{0..8})
betti(residual=X5:X)
dim residual, degree residual, degree (residual+X)
```

Text

In case $n=3$ there are 3 six secant lines:

Example

```
X=K3surfaceD11S11Ln(P4,3);
minimalBetti X
betti(X5=ideal (gens X)_{0..8})
betti(plane=X5:X)
dim plane, degree (plane+X)
tally apply(primaryDecomposition(plane+X),c->(dim c, degree radical(c+X)))
```

Text

So the plane intersects X in a plane quartic and 3 points.

The three 6-secant lines are the lines in the plane joining 2 of the three points.

Example

```
Ksquare(11,11,2)
LeBarzN6(11,11,2)
HdotK(11,11)
```

Text

Summary: X is a K3 surface with precisely $e_1=(4-n)$ exceptional lines and further exceptional curves of larger degree as in the following pattern $(e_1, e_2, e_3, \dots)$
 $(4, 0, 0, 0, 1), (3, 0, 2), (2, 2, 1), (1, 4)$

References

Popescu

SeeAlso

///

doc///

Key

K3surfaceD11S12
(K3surfaceD11S12, PolynomialRing)

Headline

construct a K3 surface of degree 11 and sectional genus 12

Usage

$X = \text{K3surfaceD11S12 } P4$

Inputs

$P4$: PolynomialRing
coordinate ring of $P4$

Outputs

X : Ideal
of a K3 surface of degree 11 and sectional genus 12

Description

Text

We construct an K3 surfaces of degree 11 and sectional genus 12

Example

```
kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
setRandomSeed("Fix decomposition of the (-1) lines");
X=K3surfaceD11S12 P4;
(d,sg)=(degree X, sectionalGenus X)
xO=chiO X
minimalBetti X
B=beti(T=tateResolutionOfSurface X)
D=canonicalDivisor X;
degree D
selfIntersectionNumber(X,D)
LeBarzN6(d,sg,xO)==9
"elapsedTime cD=primaryDecomposition D;";
"tally apply(cD,c->(dim c, degree c, betti c, selfIntersectionNumber(X,c)))";
```

Text

X has no 6-secant line, since the ideal is generated by quintics.

Thus there nine (-1) lines and a (-1) conic.

The (-1) lines group in Frobenius orbits of x, y, z elements.

References

SeeAlso

selfIntersectionNumber
LeBarzN6
canonicalDivisor
///

doc///

Key

K3surfaceD12
(K3surfaceD12, PolynomialRing)

Headline

construct a K3 surface of degree 12 and sectional genus 14

Usage

X=K3surfaceD12 P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal

of a K3 surface of degree 12 and sectional genus 14

Description

Text

We construct an K3 surfaces of degree 12 and sectional genus 14

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];  
setRandomSeed("Fix decomposition of the (-1) curves");  
X=K3surfaceD12 P4;  
(d,sg)=(degree X, sectionalGenus X)  
xO=chiO X  
minimalBetti X  
B=betti(T=tateResolutionOfSurface X)  
D=canonicalDivisor X;  
degree D  
selfIntersectionNumber(X,D)  
LeBarzN6(d,sg,xO)==10  
"elapsedTime cD=primaryDecomposition D;";  
"tally apply(cD,c->(dim c, degree c, betti c, selfIntersectionNumber(X,c)))";
```

Text

X has no 6-secant line, since the ideal is generated by quintics.

Thus there 10 (-1) lines and a (-1) quartic.

The (-1) lines group in Frobenius orbits of x,y,z elements.

References

SeeAlso

selfIntersectionNumber
LeBarzN6
canonicalDivisor
///

doc///

Key

K3surfaceD13
(K3surfaceD13, PolynomialRing)

Headline

construct a K3 surface of degree 13 and sectional genus 16

Usage

X=K3surfaceD13 P4

Inputs

P4:PolynomialRing
coordinate ring of P4

Outputs

X:Ideal

of a K3 surface of degree 13 and sectional genus 16

Description

Text

We construct an K3 surfaces of degree 13 and sectional genus 16

Example

```
kk=ZZ/nextPrime 10^3;  
P4=kk[x_0..x_4];
```

```

setRandomSeed("Fix decomposition of the (-1) curves");
X=K3surfaceD13 P4;
(d,sg)=(degree X, sectionalGenus X)
xO=chiO X
minimalBetti X
B=betti(T=tateResolutionOfSurface X)
D=canonicalDivisor X;
degree D
selfIntersectionNumber(X,D)
LeBarzN6(d,sg,xO) ==10
residual=residualInQuintics X
dim residual==0
"elapsedTime cD=primaryDecomposition D;";
"tally apply(cD,c->(dim c, degree c, betti c, selfIntersectionNumber(X,c)))";

```

Text

X has no 6-secant line, since residualInQuintics has dimension 0.

Thus there are ten (-1) lines and a (-1) septic.

The (-1) lines group in Frobenius orbits of x,y,z elements.

References

SeeAlso

```

selfIntersectionNumber
residualInQuintics
LeBarzN6
canonicalDivisor
///

```

doc///

Key

```

K3surfaceD14
(K3surfaceD14, PolynomialRing)

```

Headline

construct a K3 surface of degree 14 and sectional genus 19

Usage

```

X=K3surfaceD14 P4

```

Inputs

```

P4:PolynomialRing
coordinate ring of P4

```

Outputs

```

X:Ideal
of a K3 surface of degree 14 and sectional genus 19

```

Description

Text

We construct an K3 surfaces of degree 14 and sectional genus 19

Example

```

kk=ZZ/nextPrime 10^3;
P4=kk[x_0..x_4];
setRandomSeed("Fix decomposition of the (-1) curves");
X=K3surfaceD14 P4;
(d,sg)=(degree X, sectionalGenus X)
xO=chiO X
minimalBetti X
B=betti(T=tateResolutionOfSurface X)
D=canonicalDivisor X;
degree D
elapsedTime selfIntersectionNumber(X,D)==-15
residual=residualInQuintics X;
dim residual, degree residual
cResidual=primaryDecomposition residual;
tally apply(cResidual,c->(dim c, degree c, degree radical c, dim(c+X), degree(c+X), degree radical(c+X)))
cResidual

```

Text

There are 4 planes containing a plane sextic curve. Any line in one of the planes is a 6-secant line, and Le Barz secant formula does not apply.

Example

```

"elapsedTime cD=primaryDecomposition D;";
"tally apply(cD,c->(dim c, degree c, betti c, selfIntersectionNumber(X,c)))";

```

Text

By [Popescu?] we know that X is the blow-up of an minimal K3 surface in 11 points embedded by (Hmin;);

References

SeeAlso

selfIntersectionNumber
residualInQuintics
LeBarzN6
canonicalDivisor
///

-* Test section *-

TEST///

///

end--

-* Development section *-

restart

needsPackage "NongeneralTypeSurfacesInP4"

installPackage "NongeneralTypeSurfacesInP4"

viewHelp "NongeneralTypeSurfacesInP4"